

MASTER_CONSOLIDATED_REPORT.txt

TBD2 pipeline output - rendered to PDF

MASTER REPORT: Customer Module Unified Report

Total sections: 8

=== low_performing_entities_report_customer.txt ===

Low-Performing Metrics

S.No,Metric Id,Metric Name,Metric Value

1,ORH10103,Order Fulfillment Rate,58.16988674217188
2,CSP10105,Customer Lifetime Value (CLV),58.37308461025982
3,ITM10114,Agent Productivity,58.47035309793471
4,QSL10119,Service Improvement Rate,58.5
5,ITM10115,Interaction Cost,58.51598934043971
6,CEM10107,Customer Journey Analytics,58.51698867421719
7,CHG10106,Customer Satisfaction (CSAT) with Charging,58.53364423717522
8,CSP10114,Segment Size and Growth Rate,58.54197201865423
9,CPM10117,Time to Escalate,58.63157894736842
10,ITM10105,Customer Effort Score (CES),58.64590273151232

Low-Performing Groups

S.No,Group Id,Group Name,Group Value

1,ITM10001,Customer Interaction Management Metrics,59.28474017321785
2,CEM10001,Customer Experience Management Metrics,59.333107928047966
3,CHG10001,Charging Metrics,59.369710193204526
4,CRM10001,Customer Relationship Management Metrics,59.41264490339774
5,BIM10001,Billing and Invoice Management Metrics,59.42092938041306
6,CSR10001,Customer Support and Readiness Metrics,59.453650899400394
7,BIH10001,Billing Inquiry Handling Metrics,59.46306462358427
8,CPM10001,Customer Problem Management Metrics,59.46356095936042
9,RTN10001,Customer Retention Management Metrics,59.47203197868088
10,CFS10001,Customer Feedback and Survey Management Metrics,59.48141905396402

Low-Performing Sub-Modules

S.No,	Sub-Module Id,	Sub-Module Name,	Sub-Module Value
1,	ITM10000,	Customer Interaction Management,	59.28474017321785
2,	CEM10000,	Customer Experience Management,	59.333107928047966
3,	CHG10000,	Charging,	59.369710193204526
4,	CRM10000,	Customer Relationship Management,	59.41264490339774
5,	BIM10000,	Billing and Invoice Management,	59.42092938041306
6,	CSR10000,	Customer Support and Readiness,	59.453650899400394
7,	BIH10000,	Billing Inquiry Handling,	59.46306462358427
8,	CPM10000,	Customer Problem Management,	59.46356095936042
9,	RTN10000,	Customer Retention Management,	59.47203197868088
10,	CFS10000,	Customer Feedback and Survey Management,	59.48141905396402
Overall Module Average,CUST_001,59.48			

=== M_G_SM_correlation_report_customer.txt ===

Correlation Report (top pairs by |correlation|; full matrices are still used internally for CHAID risk binning - see chaid_risk_segmentation_report)

Metrics Correlation Summary

Total pairs examined: 60378

Pairs with |correlation| >= 0.5: 0

Mean |correlation|: 0.0147

No pairs met the threshold - showing top 150 by |correlation| anyway:

Metric 1 ID	Metric 1 Name	Metric 2 ID	Metric 2 Name	Correlation
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CCM10101	Average Response Time	CCM10115	Communication Abandonment Rate	0.0761
CSR10105	Call Abandonment Rate	CPM10116	Customer Feedback	-0.0723
ITM10110	Interaction Abandonment Rate	BEV10111	Event Volume	-0.0718
QSL10111	Uptime Percentage	BIM10115	Recurring Billing Error Rate	0.0712
QSL10113	Network Latency	CFS10116	Feedback Loop Closure Rate	-0.0711
ITM10103	Customer Satisfaction (CSAT) Score	QSL10108	Customer Satisfaction (CSAT) Score	-0.0707
BEV10114	Customer Impact Rate	RTN10106	Average Customer Tenure	0.0702
RTN10103	Customer Lifetime Value (CLV)	CII10102	Net Promoter Score (NPS)	-0.0693
CCM10104	Net Promoter Score (NPS)	CCM10117	Training Hours per Agent	-0.0685
CPM10106	Resolution Effectiveness	CCM10102	First Contact Resolution (FCR) Rate	-0.0683
CIM10108	Customer Data Update Rate	CII10102	Net Promoter Score (NPS)	0.0675
ORH10109	Order Completion Rate	CSP10115	Behavioral Patterns by Segment	0.0671
BIH10109	Agent Productivity	CBC10111	Balance Adjustment Rate	0.0670
CRM10104	Net Promoter Score (NPS)	RTN10112	Upsell and Cross-Sell Rates	0.0664

BPR10108 | Billing Cycle Time | CII10109 | Customer Interaction Volume | 0.0660
CCM10102 | First Contact Resolution (FCR) Rate | CSP10106 | Segment Profitability | -0.0660
ORH10111 | Order Entry Error Rate | BEV10118 | Customer Feedback on Billing Events | 0.0659
RTN10108 | Engagement Rate | CFS10110 | Actionable Feedback Rate | 0.0657
BPR10116 | Customer Feedback on Billing | CFS10103 | Net Promoter Score (NPS) | 0.0656
CSR10106 | Agent Productivity | CII10109 | Customer Interaction Volume | 0.0655
CSR10103 | Customer Satisfaction (CSAT) Score | RTN10116 | Retention Cost per Customer | -0.0654
ORH10105 | Order Cancellation Rate | QSL10108 | Customer Satisfaction (CSAT) Score | 0.0654
BEV10108 | Customer Notification Time | CCM10114 | Average Handling Time (AHT) | 0.0653
CIM10107 | Data Security Incidents | CII10105 | Customer Retention Rate | -0.0652
CBC10112 | Automated Balance Management Rate | CSP10105 | Customer Lifetime Value (CLV) | 0.0652
BIM10112 | Late Payment Rate | CHG10101 | Charge Accuracy Rate | -0.0643
CRM10103 | Customer Acquisition Cost (CAC) | RTN10101 | Customer Retention Rate | 0.0639
ITM10105 | Customer Effort Score (CES) | BPR10107 | Customer Satisfaction (CSAT) with Billing | -0.0635
CRM10101 | Customer Lifetime Value (CLV) | BIH10104 | Inquiry Resolution Time | -0.0635
CSR10107 | Training Hours per Agent | QSL10115 | Throughput | -0.0634
CEM10104 | Customer Churn Rate | ORH10103 | Order Fulfillment Rate | 0.0634
BIM10114 | Credit Note Issuance Rate | CSP10111 | Churn Rate by Segment | -0.0633
ITM10103 | Customer Satisfaction (CSAT) Score | CFS10107 | Average Feedback Response Time | 0.0633
CPM10115 | Agent Productivity | BEV10101 | Billing Event Accuracy Rate | 0.0632
CIM10116 | Data Quality Score | BPR10103 | On-Time Invoice Delivery Rate | 0.0628
CII10103 | Customer Lifetime Value (CLV) | CCM10106 | Communication Channel Effectiveness | -0.0628
BIM10102 | Invoice Generation Time | RTN10115 | Customer Exit Survey Results | 0.0628
CPM10113 | Service Level Agreement (SLA) Compliance | CCM10114 | Average Handling Time (AHT) | -0.0626
CPM10101 | Problem Resolution Time | BIH10115 | Inquiry Closure Rate | -0.0626
CRM10103 | Customer Acquisition Cost (CAC) | CPM10110 | Customer Retention Rate | -0.0625
CRM10119 | CRM System Utilization | RTN10103 | Customer Lifetime Value (CLV) | -0.0625
CPM10107 | Average Response Time | CHG10115 | Chargeback Rate | -0.0625
BIH10109 | Agent Productivity | CII10103 | Customer Lifetime Value (CLV) | 0.0625
ORH10102 | Order Accuracy Rate | CPM10112 | Customer Complaints Volume | 0.0624
CPM10111 | Churn Rate | CSP10101 | Segmentation Accuracy | 0.0622
QSL10103 | Mean Time to Repair (MTTR) | CII10109 | Customer Interaction Volume | -0.0622
CIM10103 | Data Timeliness | BEV10116 | Event System Uptime | -0.0621
CSR10110 | Customer Retention Rate | BIM10119 | Compliance Rate with Billing Regulations | 0.0621
QSL10108 | Customer Satisfaction (CSAT) Score | CFS10115 | Customer Participation Rate | -0.0618
QSL10114 | Packet Loss Rate | CBC10117 | Balance Management System Uptime | 0.0616
CSP10101 | Segmentation Accuracy | CSP10102 | Customer Profile Completeness | 0.0616
CRM10101 | Customer Lifetime Value (CLV) | BPR10118 | Cost per Invoice | -0.0616
QSL10102 | Service Downtime | CHG10105 | Revenue Assurance Rate | -0.0615
BIH10108 | Customer Feedback on Billing Inquiries | CCM10108 | Communication Accuracy Rate | -0.0614

ITM10114 | Agent Productivity | CBC10105 | Balance Dispute Rate | -0.0614
CIM10117 | Customer Information Accessibility | CBC10106 | Dispute Resolution Time | 0.0614
CEM10105 | Average Handling Time (AHT) | RTN10105 | Customer Satisfaction (CSAT) Score | -0.0611
CRM10101 | Customer Lifetime Value (CLV) | CFS10101 | Response Rate | -0.0610
ORH10114 | Order Payment Processing Time | CHG10108 | Dispute Resolution Time | -0.0609
ORH10112 | Order Status Updates | CFS10113 | Feedback Resolution Time | -0.0609
CRM10118 | Customer Onboarding Time | CIM10115 | Customer Data Retention Rate | -0.0608
BEV10107 | Event Verification Rate | CCM10119 | Knowledge Base Utilization | -0.0608
CPM10119 | Problem Severity Levels | CII10106 | Customer Segmentation Accuracy | -0.0607
BEV10108 | Customer Notification Time | BEV10112 | Event Audit Compliance Rate | 0.0607
BPR10116 | Customer Feedback on Billing | CSP10105 | Customer Lifetime Value (CLV) | 0.0606
CII10119 | Upsell and Cross-Sell Success Rate | CSP10103 | Segmentation Utilization Rate | 0.0606
CPM10113 | Service Level Agreement (SLA) Compliance | BEV10107 | Event Verification Rate | 0.0605
CRM10118 | Customer Onboarding Time | QSL10118 | Problem Resolution Time | -0.0605
BEV10109 | Event Dispute Rate | CFS10103 | Net Promoter Score (NPS) | 0.0603
BIH10107 | Inquiry Accuracy Rate | RTN10105 | Customer Satisfaction (CSAT) Score | 0.0602
BEV10115 | Event-Based Billing Accuracy | BEV10118 | Customer Feedback on Billing Events | -0.0600
CHG10106 | Customer Satisfaction (CSAT) with Charging | CFS10113 | Feedback Resolution Time | -0.0599
CBC10102 | Balance Update Frequency | CII10119 | Upsell and Cross-Sell Success Rate | -0.0598
ORH10104 | Order Backlog | ITM10112 | Interaction Conversion Rate | 0.0597
CHG10114 | Charging System Uptime | CFS10103 | Net Promoter Score (NPS) | -0.0597
CIM10116 | Data Quality Score | BIH10107 | Inquiry Accuracy Rate | -0.0596
CEM10114 | Customer Loyalty Programs | CCM10113 | Customer Engagement Rate | 0.0595
CRM10117 | Upsell and Cross-Sell Rates | ITM10103 | Customer Satisfaction (CSAT) Score | 0.0594
BIH10118 | Training Hours per Agent | BEV10105 | Event Correlation Success Rate | -0.0592
CEM10111 | Customer Engagement | BIH10101 | Average Response Time | -0.0592
BPR10104 | Billing Dispute Rate | BPR10106 | Automated Billing Percentage | -0.0592
BIH10110 | Repeat Inquiry Rate | RTN10112 | Upsell and Cross-Sell Rates | 0.0591
CSR10102 | First Contact Resolution (FCR) | BIH10111 | Inquiry Abandonment Rate | 0.0590
BIM10108 | Billing Cycle Time | CCM10107 | Interaction Volume by Channel | 0.0590
CSR10113 | Customer Feedback | CII10110 | Customer Feedback Collection Rate | 0.0589
CCM10111 | Repeat Contact Rate | CSP10117 | Segment-Specific Revenue | 0.0589
ITM10102 | Average Handling Time (AHT) | CPM10113 | Service Level Agreement (SLA) Compliance | -0.0588
CHG10109 | Charge Dispute Rate | CBC10108 | Overdraft Occurrence Rate | -0.0588
CBC10109 | Balance Notification Timeliness | CCM10107 | Interaction Volume by Channel | 0.0587
BIH10118 | Training Hours per Agent | CSP10101 | Segmentation Accuracy | -0.0587
CRM10109 | Customer Engagement Rate | CFS10106 | Feedback Quality Score | -0.0587
CHG10111 | Charge Consistency Rate | CFS10114 | Survey Distribution Channels | 0.0585
RTN10118 | Customer Complaints Resolution Rate | CCM10111 | Repeat Contact Rate | -0.0581
BPR10105 | Dispute Resolution Time | CFS10102 | Customer Satisfaction (CSAT) Score | -0.0579

CSR10104 | Service Level Agreement (SLA) Compliance | ITM10108 | Customer Engagement Rate | 0.0579
CHG10108 | Dispute Resolution Time | CSP10106 | Segment Profitability | -0.0579
QSL10114 | Packet Loss Rate | CSP10117 | Segment-Specific Revenue | 0.0578
ORH10106 | Order Modification Rate | CBC10116 | Customer Feedback on Balance Management | -0.0578
CRM10112 | Customer Feedback | CHG10112 | Billing Cycle Time for Charges | 0.0576
CRM10118 | Customer Onboarding Time | CSP10116 | Demographic Trends | 0.0576
ORH10106 | Order Modification Rate | CSP10119 | Cost per Segment Profile | 0.0574
BEV10107 | Event Verification Rate | CSP10117 | Segment-Specific Revenue | -0.0574
CSP10107 | Customer Engagement Rate by Segment | CFS10106 | Feedback Quality Score | 0.0573
BIM10119 | Compliance Rate with Billing Regulations | CCM10117 | Training Hours per Agent | 0.0573
ITM10119 | Customer Retention Rate | BIM10114 | Credit Note Issuance Rate | 0.0573
BEV10117 | Event Integration Success Rate | CCM10111 | Repeat Contact Rate | -0.0573
ORH10105 | Order Cancellation Rate | CIM10111 | Customer Privacy Compliance | 0.0572
BIH10105 | Billing Inquiry Volume | CFS10113 | Feedback Resolution Time | -0.0572
CIM10118 | Data Anonymization Rate | BIM10103 | On-Time Invoice Delivery Rate | 0.0572
ORH10111 | Order Entry Error Rate | QSL10113 | Network Latency | 0.0570
CRM10116 | Customer Touchpoint Effectiveness | QSL10109 | Net Promoter Score (NPS) | 0.0570
CEM10106 | First Contact Resolution (FCR) | CBC10115 | Balance Monitoring Coverage | 0.0569
CHG10115 | Chargeback Rate | RTN10106 | Average Customer Tenure | -0.0568
CCM10112 | Communication Quality Score | CFS10105 | Survey Completion Rate | 0.0568
CIM10109 | Data Usage Rate | BIM10112 | Late Payment Rate | -0.0568
QSL10105 | SLA Compliance Rate | BPR10115 | Recurring Billing Error Rate | 0.0568
BIH10103 | Customer Satisfaction (CSAT) with Billing Inquiries | CCM10116 | Customer Retention Rate | -0.0567
QSL10110 | Service Quality Index (SQI) | CII10113 | Customer Data Integration Rate | -0.0567
CCM10103 | Customer Satisfaction (CSAT) with Communication | CCM10112 | Communication Quality Score | -0.0567
BEV10117 | Event Integration Success Rate | BEV10118 | Customer Feedback on Billing Events | -0.0567
CFS10102 | Customer Satisfaction (CSAT) Score | CFS10118 | Survey Design Effectiveness | 0.0566
CEM10114 | Customer Loyalty Programs | QSL10109 | Net Promoter Score (NPS) | 0.0566
CSR10110 | Customer Retention Rate | BEV10109 | Event Dispute Rate | 0.0566
ITM10115 | Interaction Cost | BIH10109 | Agent Productivity | 0.0566
QSL10103 | Mean Time to Repair (MTTR) | CFS10106 | Feedback Quality Score | 0.0565
QSL10102 | Service Downtime | BIM10103 | On-Time Invoice Delivery Rate | 0.0565
CHG10115 | Chargeback Rate | CBC10117 | Balance Management System Uptime | 0.0564
BIH10104 | Inquiry Resolution Time | RTN10112 | Upsell and Cross-Sell Rates | -0.0563
CIM10107 | Data Security Incidents | CFS10108 | Sentiment Analysis | -0.0563
BIH10110 | Repeat Inquiry Rate | CSP10113 | Customer Satisfaction (CSAT) Score by Segment | 0.0563
CRM10109 | Customer Engagement Rate | ITM10105 | Customer Effort Score (CES) | 0.0563
CHG10105 | Revenue Assurance Rate | CSP10115 | Behavioral Patterns by Segment | -0.0563
BIM10105 | Dispute Resolution Time | CII10103 | Customer Lifetime Value (CLV) | -0.0562

BEV10106 | Real-Time Event Processing Rate | BEV10119 | Cost per Event Processed | 0.0562
QSL10106 | Incident Response Time | BIM10101 | Billing Accuracy Rate | 0.0560
CBC10114 | Balance Data Integrity | CFS10119 | Cost per Survey Response | -0.0560
CRM10115 | Customer Segmentation | CII10118 | Conversion Rate | 0.0559
ITM10109 | Interaction Quality Score | QSL10102 | Service Downtime | 0.0559
ITM10103 | Customer Satisfaction (CSAT) Score | CIM10109 | Data Usage Rate | -0.0558
CSR10112 | Escalation Rate | CFS10112 | Follow-up Rate | -0.0558
CSP10105 | Customer Lifetime Value (CLV) | CSP10113 | Customer Satisfaction (CSAT) Score by Segment | -0.0558
BPR10112 | Late Payment Rate | BPR10117 | Billing System Uptime | 0.0558
QSL10104 | Mean Time Between Failures (MTBF) | RTN10101 | Customer Retention Rate | 0.0556
CPM10102 | First Contact Resolution (FCR) Rate | QSL10111 | Uptime Percentage | -0.0556
ORH10106 | Order Modification Rate | BPR10106 | Automated Billing Percentage | 0.0556
QSL10105 | SLA Compliance Rate | BPR10107 | Customer Satisfaction (CSAT) with Billing | -0.0555
CRM10109 | Customer Engagement Rate | CHG10107 | Charge Reversal Rate | 0.0554
CIM10110 | Data Redundancy | CFS10110 | Actionable Feedback Rate | -0.0554
ITM10118 | Interaction Channel Preference | CPM10119 | Problem Severity Levels | 0.0554
CSR10109 | Ticket Volume | CSP10114 | Segment Size and Growth Rate | 0.0553

Groups Correlation Summary

Total pairs examined: 171
Pairs with |correlation| >= 0.5: 0
Mean |correlation|: 0.0142

No pairs met the threshold - showing top 150 by |correlation| anyway:

Metric 1 ID	Metric 1 Name	Metric 2 ID	Metric 2 Name	Correlation
BEV10001	Manage Billing Events Metrics	CHG10001	Charging Metrics	-0.0521
CIM10001	Customer Information Management Metrics	CCM10001	Customer Communication Management Metrics	0.0431
ORH10001	Order Handling Metrics	CPM10001	Customer Problem Management Metrics	0.0424
ITM10001	Customer Interaction Management Metrics	CSP10001	Customer Segmentation and Profiling Metrics	0.0395
CSR10001	Customer Support and Readiness Metrics	CSP10001	Customer Segmentation and Profiling Metrics	-0.0382
CPM10001	Customer Problem Management Metrics	BEV10001	Manage Billing Events Metrics	0.0374
BIM10001	Billing and Invoice Management Metrics	CSP10001	Customer Segmentation and Profiling Metrics	-0.0373
CSR10001	Customer Support and Readiness Metrics	BIM10001	Billing and Invoice Management Metrics	0.0360
CSR10001	Customer Support and Readiness Metrics	ITM10001	Customer Interaction Management Metrics	0.0356
CSR10001	Customer Support and Readiness Metrics	CEM10001	Customer Experience Management Metrics	-0.0355

CIM10001 | Customer Information Management Metrics | RTN10001 | Customer Retention Management Metrics | 0.0344

CEM10001 | Customer Experience Management Metrics | BIM10001 | Billing and Invoice Management Metrics | -0.0327

QSL10001 | Customer QoS/SLA Management Metrics | CII10001 | Customer Insight and Intelligence Metrics | -0.0325

CPM10001 | Customer Problem Management Metrics | BIH10001 | Billing Inquiry Handling Metrics | -0.0316

CPM10001 | Customer Problem Management Metrics | CCM10001 | Customer Communication Management Metrics | -0.0312

ORH10001 | Order Handling Metrics | CRM10001 | Customer Relationship Management Metrics | -0.0308

ORH10001 | Order Handling Metrics | BPR10001 | Bill Payments and Receivables Management Metrics | -0.0306

BIM10001 | Billing and Invoice Management Metrics | BIH10001 | Billing Inquiry Handling Metrics | 0.0305

CHG10001 | Charging Metrics | CSP10001 | Customer Segmentation and Profiling Metrics | 0.0300

CRM10001 | Customer Relationship Management Metrics | CSP10001 | Customer Segmentation and Profiling Metrics | 0.0298

CPM10001 | Customer Problem Management Metrics | CFS10001 | Customer Feedback and Survey Management Metrics | -0.0283

CSR10001 | Customer Support and Readiness Metrics | CFS10001 | Customer Feedback and Survey Management Metrics | 0.0281

ORH10001 | Order Handling Metrics | QSL10001 | Customer QoS/SLA Management Metrics | -0.0277

CIM10001 | Customer Information Management Metrics | BEV10001 | Manage Billing Events Metrics | 0.0277

BPR10001 | Bill Payments and Receivables Management Metrics | CBC10001 | Manage Balances Metrics | -0.0277

BEV10001 | Manage Billing Events Metrics | CBC10001 | Manage Balances Metrics | -0.0271

BIH10001 | Billing Inquiry Handling Metrics | CII10001 | Customer Insight and Intelligence Metrics | -0.0270

CSR10001 | Customer Support and Readiness Metrics | QSL10001 | Customer QoS/SLA Management Metrics | -0.0267

CRM10001 | Customer Relationship Management Metrics | ITM10001 | Customer Interaction Management Metrics | -0.0266

CHG10001 | Charging Metrics | CBC10001 | Manage Balances Metrics | -0.0258

CSR10001 | Customer Support and Readiness Metrics | CCM10001 | Customer Communication Management Metrics | 0.0255

ORH10001 | Order Handling Metrics | BEV10001 | Manage Billing Events Metrics | 0.0253

CPM10001 | Customer Problem Management Metrics | BPR10001 | Bill Payments and Receivables Management Metrics | 0.0252

CRM10001 | Customer Relationship Management Metrics | CHG10001 | Charging Metrics | 0.0252

CHG10001 | Charging Metrics | CCM10001 | Customer Communication Management Metrics | -0.0250

BIM10001 | Billing and Invoice Management Metrics | CFS10001 | Customer Feedback and Survey Management Metrics | 0.0248

BPR10001 | Bill Payments and Receivables Management Metrics | BIH10001 | Billing Inquiry Handling Metrics | -0.0239

CSR10001 | Customer Support and Readiness Metrics | CHG10001 | Charging Metrics | 0.0237

CRM10001 | Customer Relationship Management Metrics | CIM10001 | Customer Information Management Metrics | -0.0236

QSL10001 | Customer QoS/SLA Management Metrics | CBC10001 | Manage Balances Metrics | 0.0222

RTN10001 | Customer Retention Management Metrics | CFS10001 | Customer Feedback and Survey Management Metrics | -0.0221

ORH10001 | Order Handling Metrics | ITM10001 | Customer Interaction Management Metrics | 0.0221

BIM10001 | Billing and Invoice Management Metrics | RTN10001 | Customer Retention Management Metrics | 0.0217

CSR10001 | Customer Support and Readiness Metrics | CII10001 | Customer Insight and Intelligence Metrics | -0.0213

CEM10001 | Customer Experience Management Metrics | CII10001 | Customer Insight and Intelligence Metrics | -0.0211

CRM10001 | Customer Relationship Management Metrics | BIH10001 | Billing Inquiry Handling Metrics | 0.0203

CSR10001 | Customer Support and Readiness Metrics | CPM10001 | Customer Problem Management Metrics | -0.0203

ORH10001 | Order Handling Metrics | CII10001 | Customer Insight and Intelligence Metrics | 0.0200

CEM10001 | Customer Experience Management Metrics | CCM10001 | Customer Communication Management Metrics | -0.0198

ORH10001 | Order Handling Metrics | CFS10001 | Customer Feedback and Survey Management Metrics | 0.0193

CEM10001 | Customer Experience Management Metrics | CSP10001 | Customer Segmentation and Profiling Metrics | 0.0192

CPM10001 | Customer Problem Management Metrics | QSL10001 | Customer QoS/SLA Management Metrics | 0.0191

CIM10001 | Customer Information Management Metrics | CPM10001 | Customer Problem Management Metrics | 0.0189

BPR10001 | Bill Payments and Receivables Management Metrics | CCM10001 | Customer Communication Management Metrics | 0.0187

CRM10001 | Customer Relationship Management Metrics | CCM10001 | Customer Communication Management Metrics | -0.0186

CBC10001 | Manage Balances Metrics | CSP10001 | Customer Segmentation and Profiling Metrics | -0.0185

CIM10001 | Customer Information Management Metrics | CSP10001 | Customer Segmentation and Profiling Metrics | -0.0184

CIM10001 | Customer Information Management Metrics | BIM10001 | Billing and Invoice Management Metrics | 0.0181

BIH10001 | Billing Inquiry Handling Metrics | CBC10001 | Manage Balances Metrics | -0.0173

BIM10001 | Billing and Invoice Management Metrics | BEV10001 | Manage Billing Events Metrics | -0.0171

BEV10001 | Manage Billing Events Metrics | CII10001 | Customer Insight and Intelligence Metrics | 0.0168

CHG10001 | Charging Metrics | CFS10001 | Customer Feedback and Survey Management Metrics | -0.0166

CPM10001 | Customer Problem Management Metrics | BIM10001 | Billing and Invoice Management Metrics | -0.0165

CRM10001 | Customer Relationship Management Metrics | CPM10001 | Customer Problem Management Metrics | 0.0164

ITM10001 | Customer Interaction Management Metrics | BIM10001 | Billing and Invoice Management Metrics | 0.0156

CEM10001 | Customer Experience Management Metrics | CRM10001 | Customer Relationship Management Metrics | -0.0156

CRM10001 | Customer Relationship Management Metrics | BPR10001 | Bill Payments and Receivables Management Metrics | 0.0147

ITM10001 | Customer Interaction Management Metrics | CFS10001 | Customer Feedback and Survey Management Metrics | -0.0147

ORH10001 | Order Handling Metrics | CHG10001 | Charging Metrics | -0.0144

CEM10001 | Customer Experience Management Metrics | ORH10001 | Order Handling Metrics | -0.0144

CEM10001 | Customer Experience Management Metrics | BIH10001 | Billing Inquiry Handling Metrics | -0.0143

ORH10001 | Order Handling Metrics | CIM10001 | Customer Information Management Metrics | -0.0134

CEM10001 | Customer Experience Management Metrics | CPM10001 | Customer Problem Management Metrics | 0.0134

CSR10001 | Customer Support and Readiness Metrics | ORH10001 | Order Handling Metrics | -0.0131

CRM10001 | Customer Relationship Management Metrics | BEV10001 | Manage Billing Events Metrics | -0.0131

ORH10001 | Order Handling Metrics | RTN10001 | Customer Retention Management Metrics | 0.0129

RTN10001 | Customer Retention Management Metrics | CII10001 | Customer Insight and Intelligence Metrics | 0.0125

ORH10001 | Order Handling Metrics | BIM10001 | Billing and Invoice Management Metrics | 0.0124

ITM10001 | Customer Interaction Management Metrics | CCM10001 | Customer Communication Management Metrics | 0.0124

CPM10001 | Customer Problem Management Metrics | CII10001 | Customer Insight and Intelligence Metrics | 0.0123

BIM10001 | Billing and Invoice Management Metrics | CCM10001 | Customer Communication Management Metrics | -0.0119

ORH10001 | Order Handling Metrics | CCM10001 | Customer Communication Management Metrics | 0.0116

QSL10001 | Customer QoS/SLA Management Metrics | BIH10001 | Billing Inquiry Handling Metrics | 0.0113

CCM10001 | Customer Communication Management Metrics | CSP10001 | Customer Segmentation and Profiling Metrics | -0.0113

CSR10001 | Customer Support and Readiness Metrics | CBC10001 | Manage Balances Metrics | -0.0112

CII10001 | Customer Insight and Intelligence Metrics | CSP10001 | Customer Segmentation and Profiling Metrics | 0.0111

CEM10001 | Customer Experience Management Metrics | RTN10001 | Customer Retention Management Metrics | -0.0111

CEM10001 | Customer Experience Management Metrics | BEV10001 | Manage Billing Events Metrics | 0.0110

QSL10001 | Customer QoS/SLA Management Metrics | BIM10001 | Billing and Invoice Management Metrics | 0.0108

CSR10001 | Customer Support and Readiness Metrics | CRM10001 | Customer Relationship Management Metrics | 0.0105

ORH10001 | Order Handling Metrics | CBC10001 | Manage Balances Metrics | -0.0105

CEM10001 | Customer Experience Management Metrics | BPR10001 | Bill Payments and Receivables Management Metrics | 0.0102

BPR10001 | Bill Payments and Receivables Management Metrics | CII10001 | Customer Insight and Intelligence Metrics | 0.0101

ITM10001 | Customer Interaction Management Metrics | CIM10001 | Customer Information Management Metrics | 0.0100

ORH10001 | Order Handling Metrics | CSP10001 | Customer Segmentation and Profiling Metrics | 0.0097
BPR10001 | Bill Payments and Receivables Management Metrics | BEV10001 | Manage Billing Events Metrics | -0.0097
RTN10001 | Customer Retention Management Metrics | CCM10001 | Customer Communication Management Metrics | -0.0096
CSR10001 | Customer Support and Readiness Metrics | BIH10001 | Billing Inquiry Handling Metrics | 0.0096
CEM10001 | Customer Experience Management Metrics | CBC10001 | Manage Balances Metrics | 0.0094
CSR10001 | Customer Support and Readiness Metrics | RTN10001 | Customer Retention Management Metrics | -0.0093
RTN10001 | Customer Retention Management Metrics | CSP10001 | Customer Segmentation and Profiling Metrics | 0.0092
CIM10001 | Customer Information Management Metrics | QSL10001 | Customer QoS/SLA Management Metrics | 0.0092
CSP10001 | Customer Segmentation and Profiling Metrics | CFS10001 | Customer Feedback and Survey Management Metrics | -0.0090
BPR10001 | Bill Payments and Receivables Management Metrics | CSP10001 | Customer Segmentation and Profiling Metrics | 0.0090
ITM10001 | Customer Interaction Management Metrics | CPM10001 | Customer Problem Management Metrics | -0.0089
CPM10001 | Customer Problem Management Metrics | CHG10001 | Charging Metrics | -0.0085
CII10001 | Customer Insight and Intelligence Metrics | CFS10001 | Customer Feedback and Survey Management Metrics | -0.0084
ITM10001 | Customer Interaction Management Metrics | RTN10001 | Customer Retention Management Metrics | 0.0080
CRM10001 | Customer Relationship Management Metrics | RTN10001 | Customer Retention Management Metrics | -0.0076
BIH10001 | Billing Inquiry Handling Metrics | CCM10001 | Customer Communication Management Metrics | -0.0074
CBC10001 | Manage Balances Metrics | CCM10001 | Customer Communication Management Metrics | -0.0074
BIH10001 | Billing Inquiry Handling Metrics | BEV10001 | Manage Billing Events Metrics | 0.0073
ITM10001 | Customer Interaction Management Metrics | CHG10001 | Charging Metrics | 0.0072
CIM10001 | Customer Information Management Metrics | CHG10001 | Charging Metrics | -0.0072
ITM10001 | Customer Interaction Management Metrics | QSL10001 | Customer QoS/SLA Management Metrics | 0.0072
CRM10001 | Customer Relationship Management Metrics | BIM10001 | Billing and Invoice Management Metrics | -0.0071
CCM10001 | Customer Communication Management Metrics | CFS10001 | Customer Feedback and Survey Management Metrics | 0.0071
BIM10001 | Billing and Invoice Management Metrics | CBC10001 | Manage Balances Metrics | 0.0069
ORH10001 | Order Handling Metrics | BIH10001 | Billing Inquiry Handling Metrics | 0.0069
BPR10001 | Bill Payments and Receivables Management Metrics | CFS10001 | Customer Feedback and Survey Management Metrics | 0.0068
CEM10001 | Customer Experience Management Metrics | CHG10001 | Charging Metrics | -0.0067
BPR10001 | Bill Payments and Receivables Management Metrics | CHG10001 | Charging Metrics | -0.0065

CRM10001 | Customer Relationship Management Metrics | CFS10001 | Customer Feedback and Survey Management Metrics | -0.0063

CRM10001 | Customer Relationship Management Metrics | CBC10001 | Manage Balances Metrics | 0.0060

CRM10001 | Customer Relationship Management Metrics | CII10001 | Customer Insight and Intelligence Metrics | -0.0058

BEV10001 | Manage Billing Events Metrics | CFS10001 | Customer Feedback and Survey Management Metrics | -0.0056

CPM10001 | Customer Problem Management Metrics | RTN10001 | Customer Retention Management Metrics | -0.0053

BEV10001 | Manage Billing Events Metrics | RTN10001 | Customer Retention Management Metrics | 0.0053

CBC10001 | Manage Balances Metrics | CII10001 | Customer Insight and Intelligence Metrics | -0.0052

CEM10001 | Customer Experience Management Metrics | ITM10001 | Customer Interaction Management Metrics | 0.0051

CEM10001 | Customer Experience Management Metrics | CFS10001 | Customer Feedback and Survey Management Metrics | -0.0050

QSL10001 | Customer QoS/SLA Management Metrics | BEV10001 | Manage Billing Events Metrics | -0.0048

ITM10001 | Customer Interaction Management Metrics | CII10001 | Customer Insight and Intelligence Metrics | -0.0047

BPR10001 | Bill Payments and Receivables Management Metrics | RTN10001 | Customer Retention Management Metrics | -0.0045

CIM10001 | Customer Information Management Metrics | BPR10001 | Bill Payments and Receivables Management Metrics | 0.0044

CIM10001 | Customer Information Management Metrics | BIH10001 | Billing Inquiry Handling Metrics | 0.0042

CSR10001 | Customer Support and Readiness Metrics | BEV10001 | Manage Billing Events Metrics | -0.0039

QSL10001 | Customer QoS/SLA Management Metrics | CSP10001 | Customer Segmentation and Profiling Metrics | 0.0038

CBC10001 | Manage Balances Metrics | CFS10001 | Customer Feedback and Survey Management Metrics | -0.0037

QSL10001 | Customer QoS/SLA Management Metrics | RTN10001 | Customer Retention Management Metrics | 0.0037

CIM10001 | Customer Information Management Metrics | CII10001 | Customer Insight and Intelligence Metrics | 0.0036

BIH10001 | Billing Inquiry Handling Metrics | CFS10001 | Customer Feedback and Survey Management Metrics | 0.0036

CIM10001 | Customer Information Management Metrics | CFS10001 | Customer Feedback and Survey Management Metrics | -0.0036

ITM10001 | Customer Interaction Management Metrics | BEV10001 | Manage Billing Events Metrics | -0.0035

QSL10001 | Customer QoS/SLA Management Metrics | CHG10001 | Charging Metrics | -0.0035

BIM10001 | Billing and Invoice Management Metrics | CII10001 | Customer Insight and Intelligence Metrics | -0.0030

CEM10001 | Customer Experience Management Metrics | CIM10001 | Customer Information Management Metrics | -0.0029

BIM10001 | Billing and Invoice Management Metrics | BPR10001 | Bill Payments and Receivables Management Metrics | 0.0029

CIM10001 | Customer Information Management Metrics | CBC10001 | Manage Balances Metrics | -0.0028
CII10001 | Customer Insight and Intelligence Metrics | CCM10001 | Customer Communication Management Metrics | -0.0026

Sub_Modules Correlation Summary

Total pairs examined: 171

Pairs with |correlation| >= 0.5: 0

Mean |correlation|: 0.0142

No pairs met the threshold - showing top 150 by |correlation| anyway:

Metric 1 ID | Metric 1 Name | Metric 2 ID | Metric 2 Name | Correlation

BEV10000 | Manage Billing Events | CHG10000 | Charging | -0.0521
CIM10000 | Customer Information Management | CCM10000 | Customer Communication Management | 0.0431
ORH10000 | Order Handling | CPM10000 | Customer Problem Management | 0.0424
ITM10000 | Customer Interaction Management | CSP10000 | Customer Segmentation and Profiling | 0.0395
CSR10000 | Customer Support and Readiness | CSP10000 | Customer Segmentation and Profiling | -0.0382
CPM10000 | Customer Problem Management | BEV10000 | Manage Billing Events | 0.0374
BIM10000 | Billing and Invoice Management | CSP10000 | Customer Segmentation and Profiling | -0.0373
CSR10000 | Customer Support and Readiness | BIM10000 | Billing and Invoice Management | 0.0360
CSR10000 | Customer Support and Readiness | ITM10000 | Customer Interaction Management | 0.0356
CSR10000 | Customer Support and Readiness | CEM10000 | Customer Experience Management | -0.0355
CIM10000 | Customer Information Management | RTN10000 | Customer Retention Management | 0.0344
CEM10000 | Customer Experience Management | BIM10000 | Billing and Invoice Management | -0.0327
QSL10000 | Customer QoS/SLA Management | CII10000 | Customer Insight and Intelligence | -0.0325
CPM10000 | Customer Problem Management | BIH10000 | Billing Inquiry Handling | -0.0316
CPM10000 | Customer Problem Management | CCM10000 | Customer Communication Management | -0.0312
ORH10000 | Order Handling | CRM10000 | Customer Relationship Management | -0.0308
ORH10000 | Order Handling | BPR10000 | Bill Payments and Receivables Management | -0.0306
BIM10000 | Billing and Invoice Management | BIH10000 | Billing Inquiry Handling | 0.0305
CHG10000 | Charging | CSP10000 | Customer Segmentation and Profiling | 0.0300
CRM10000 | Customer Relationship Management | CSP10000 | Customer Segmentation and Profiling | 0.0298
CPM10000 | Customer Problem Management | CFS10000 | Customer Feedback and Survey Management | -0.0283
CSR10000 | Customer Support and Readiness | CFS10000 | Customer Feedback and Survey Management | 0.0281
ORH10000 | Order Handling | QSL10000 | Customer QoS/SLA Management | -0.0277
CIM10000 | Customer Information Management | BEV10000 | Manage Billing Events | 0.0277
BPR10000 | Bill Payments and Receivables Management | CBC10000 | Manage Balances | -0.0277
BEV10000 | Manage Billing Events | CBC10000 | Manage Balances | -0.0271
BIH10000 | Billing Inquiry Handling | CII10000 | Customer Insight and Intelligence | -0.0270
CSR10000 | Customer Support and Readiness | QSL10000 | Customer QoS/SLA Management | -0.0267
CRM10000 | Customer Relationship Management | ITM10000 | Customer Interaction Management | -0.0266

CHG10000 | Charging | CBC10000 | Manage Balances | -0.0258
CSR10000 | Customer Support and Readiness | CCM10000 | Customer Communication Management | 0.0255
ORH10000 | Order Handling | BEV10000 | Manage Billing Events | 0.0253
CPM10000 | Customer Problem Management | BPR10000 | Bill Payments and Receivables Management | 0.0252
CRM10000 | Customer Relationship Management | CHG10000 | Charging | 0.0252
CHG10000 | Charging | CCM10000 | Customer Communication Management | -0.0250
BIM10000 | Billing and Invoice Management | CFS10000 | Customer Feedback and Survey Management | 0.0248
BPR10000 | Bill Payments and Receivables Management | BIH10000 | Billing Inquiry Handling | -0.0239
CSR10000 | Customer Support and Readiness | CHG10000 | Charging | 0.0237
CRM10000 | Customer Relationship Management | CIM10000 | Customer Information Management | -0.0236
QSL10000 | Customer QoS/SLA Management | CBC10000 | Manage Balances | 0.0222
RTN10000 | Customer Retention Management | CFS10000 | Customer Feedback and Survey Management |
-0.0221
ORH10000 | Order Handling | ITM10000 | Customer Interaction Management | 0.0221
BIM10000 | Billing and Invoice Management | RTN10000 | Customer Retention Management | 0.0217
CSR10000 | Customer Support and Readiness | CII10000 | Customer Insight and Intelligence | -0.0213
CEM10000 | Customer Experience Management | CII10000 | Customer Insight and Intelligence | -0.0211
CRM10000 | Customer Relationship Management | BIH10000 | Billing Inquiry Handling | 0.0203
CSR10000 | Customer Support and Readiness | CPM10000 | Customer Problem Management | -0.0203
ORH10000 | Order Handling | CII10000 | Customer Insight and Intelligence | 0.0200
CEM10000 | Customer Experience Management | CCM10000 | Customer Communication Management | -0.0198
ORH10000 | Order Handling | CFS10000 | Customer Feedback and Survey Management | 0.0193
CEM10000 | Customer Experience Management | CSP10000 | Customer Segmentation and Profiling | 0.0192
CPM10000 | Customer Problem Management | QSL10000 | Customer QoS/SLA Management | 0.0191
CIM10000 | Customer Information Management | CPM10000 | Customer Problem Management | 0.0189
BPR10000 | Bill Payments and Receivables Management | CCM10000 | Customer Communication Management |
0.0187
CRM10000 | Customer Relationship Management | CCM10000 | Customer Communication Management | -0.0186
CBC10000 | Manage Balances | CSP10000 | Customer Segmentation and Profiling | -0.0185
CIM10000 | Customer Information Management | CSP10000 | Customer Segmentation and Profiling | -0.0184
CIM10000 | Customer Information Management | BIM10000 | Billing and Invoice Management | 0.0181
BIH10000 | Billing Inquiry Handling | CBC10000 | Manage Balances | -0.0173
BIM10000 | Billing and Invoice Management | BEV10000 | Manage Billing Events | -0.0171
BEV10000 | Manage Billing Events | CII10000 | Customer Insight and Intelligence | 0.0168
CHG10000 | Charging | CFS10000 | Customer Feedback and Survey Management | -0.0166
CPM10000 | Customer Problem Management | BIM10000 | Billing and Invoice Management | -0.0165
CRM10000 | Customer Relationship Management | CPM10000 | Customer Problem Management | 0.0164
ITM10000 | Customer Interaction Management | BIM10000 | Billing and Invoice Management | 0.0156
CEM10000 | Customer Experience Management | CRM10000 | Customer Relationship Management | -0.0156

CRM10000 | Customer Relationship Management | BPR10000 | Bill Payments and Receivables Management | 0.0147

ITM10000 | Customer Interaction Management | CFS10000 | Customer Feedback and Survey Management | -0.0147

ORH10000 | Order Handling | CHG10000 | Charging | -0.0144

CEM10000 | Customer Experience Management | ORH10000 | Order Handling | -0.0144

CEM10000 | Customer Experience Management | BIH10000 | Billing Inquiry Handling | -0.0143

ORH10000 | Order Handling | CIM10000 | Customer Information Management | -0.0134

CEM10000 | Customer Experience Management | CPM10000 | Customer Problem Management | 0.0134

CSR10000 | Customer Support and Readiness | ORH10000 | Order Handling | -0.0131

CRM10000 | Customer Relationship Management | BEV10000 | Manage Billing Events | -0.0131

ORH10000 | Order Handling | RTN10000 | Customer Retention Management | 0.0129

RTN10000 | Customer Retention Management | CII10000 | Customer Insight and Intelligence | 0.0125

ORH10000 | Order Handling | BIM10000 | Billing and Invoice Management | 0.0124

ITM10000 | Customer Interaction Management | CCM10000 | Customer Communication Management | 0.0124

CPM10000 | Customer Problem Management | CII10000 | Customer Insight and Intelligence | 0.0123

BIM10000 | Billing and Invoice Management | CCM10000 | Customer Communication Management | -0.0119

ORH10000 | Order Handling | CCM10000 | Customer Communication Management | 0.0116

QSL10000 | Customer QoS/SLA Management | BIH10000 | Billing Inquiry Handling | 0.0113

CCM10000 | Customer Communication Management | CSP10000 | Customer Segmentation and Profiling | -0.0113

CSR10000 | Customer Support and Readiness | CBC10000 | Manage Balances | -0.0112

CII10000 | Customer Insight and Intelligence | CSP10000 | Customer Segmentation and Profiling | 0.0111

CEM10000 | Customer Experience Management | RTN10000 | Customer Retention Management | -0.0111

CEM10000 | Customer Experience Management | BEV10000 | Manage Billing Events | 0.0110

QSL10000 | Customer QoS/SLA Management | BIM10000 | Billing and Invoice Management | 0.0108

CSR10000 | Customer Support and Readiness | CRM10000 | Customer Relationship Management | 0.0105

ORH10000 | Order Handling | CBC10000 | Manage Balances | -0.0105

CEM10000 | Customer Experience Management | BPR10000 | Bill Payments and Receivables Management | 0.0102

BPR10000 | Bill Payments and Receivables Management | CII10000 | Customer Insight and Intelligence | 0.0101

ITM10000 | Customer Interaction Management | CIM10000 | Customer Information Management | 0.0100

ORH10000 | Order Handling | CSP10000 | Customer Segmentation and Profiling | 0.0097

BPR10000 | Bill Payments and Receivables Management | BEV10000 | Manage Billing Events | -0.0097

RTN10000 | Customer Retention Management | CCM10000 | Customer Communication Management | -0.0096

CSR10000 | Customer Support and Readiness | BIH10000 | Billing Inquiry Handling | 0.0096

CEM10000 | Customer Experience Management | CBC10000 | Manage Balances | 0.0094

CSR10000 | Customer Support and Readiness | RTN10000 | Customer Retention Management | -0.0093

RTN10000 | Customer Retention Management | CSP10000 | Customer Segmentation and Profiling | 0.0092

CIM10000 | Customer Information Management | QSL10000 | Customer QoS/SLA Management | 0.0092

CSP10000 | Customer Segmentation and Profiling | CFS10000 | Customer Feedback and Survey Management | -0.0090

BPR10000 | Bill Payments and Receivables Management | CSP10000 | Customer Segmentation and Profiling | 0.0090

ITM10000 | Customer Interaction Management | CPM10000 | Customer Problem Management | -0.0089

CPM10000 | Customer Problem Management | CHG10000 | Charging | -0.0085

CII10000 | Customer Insight and Intelligence | CFS10000 | Customer Feedback and Survey Management | -0.0084

ITM10000 | Customer Interaction Management | RTN10000 | Customer Retention Management | 0.0080

CRM10000 | Customer Relationship Management | RTN10000 | Customer Retention Management | -0.0076

BIH10000 | Billing Inquiry Handling | CCM10000 | Customer Communication Management | -0.0074

CBC10000 | Manage Balances | CCM10000 | Customer Communication Management | -0.0074

BIH10000 | Billing Inquiry Handling | BEV10000 | Manage Billing Events | 0.0073

ITM10000 | Customer Interaction Management | CHG10000 | Charging | 0.0072

CIM10000 | Customer Information Management | CHG10000 | Charging | -0.0072

ITM10000 | Customer Interaction Management | QSL10000 | Customer QoS/SLA Management | 0.0072

CRM10000 | Customer Relationship Management | BIM10000 | Billing and Invoice Management | -0.0071

CCM10000 | Customer Communication Management | CFS10000 | Customer Feedback and Survey Management | 0.0071

BIM10000 | Billing and Invoice Management | CBC10000 | Manage Balances | 0.0069

ORH10000 | Order Handling | BIH10000 | Billing Inquiry Handling | 0.0069

BPR10000 | Bill Payments and Receivables Management | CFS10000 | Customer Feedback and Survey Management | 0.0068

CEM10000 | Customer Experience Management | CHG10000 | Charging | -0.0067

BPR10000 | Bill Payments and Receivables Management | CHG10000 | Charging | -0.0065

CRM10000 | Customer Relationship Management | CFS10000 | Customer Feedback and Survey Management | -0.0063

CRM10000 | Customer Relationship Management | CBC10000 | Manage Balances | 0.0060

CRM10000 | Customer Relationship Management | CII10000 | Customer Insight and Intelligence | -0.0058

BEV10000 | Manage Billing Events | CFS10000 | Customer Feedback and Survey Management | -0.0056

CPM10000 | Customer Problem Management | RTN10000 | Customer Retention Management | -0.0053

BEV10000 | Manage Billing Events | RTN10000 | Customer Retention Management | 0.0053

CBC10000 | Manage Balances | CII10000 | Customer Insight and Intelligence | -0.0052

CEM10000 | Customer Experience Management | ITM10000 | Customer Interaction Management | 0.0051

CEM10000 | Customer Experience Management | CFS10000 | Customer Feedback and Survey Management | -0.0050

QSL10000 | Customer QoS/SLA Management | BEV10000 | Manage Billing Events | -0.0048

ITM10000 | Customer Interaction Management | CII10000 | Customer Insight and Intelligence | -0.0047

BPR10000 | Bill Payments and Receivables Management | RTN10000 | Customer Retention Management | -0.0045

CIM10000 | Customer Information Management | BPR10000 | Bill Payments and Receivables Management | 0.0044

CIM10000 | Customer Information Management | BIH10000 | Billing Inquiry Handling | 0.0042

CSR10000 | Customer Support and Readiness | BEV10000 | Manage Billing Events | -0.0039

QSL10000 | Customer QoS/SLA Management | CSP10000 | Customer Segmentation and Profiling | 0.0038

CBC10000 | Manage Balances | CFS10000 | Customer Feedback and Survey Management | -0.0037
QSL10000 | Customer QoS/SLA Management | RTN10000 | Customer Retention Management | 0.0037
CIM10000 | Customer Information Management | CII10000 | Customer Insight and Intelligence | 0.0036
BIH10000 | Billing Inquiry Handling | CFS10000 | Customer Feedback and Survey Management | 0.0036
CIM10000 | Customer Information Management | CFS10000 | Customer Feedback and Survey Management | -0.0036
ITM10000 | Customer Interaction Management | BEV10000 | Manage Billing Events | -0.0035
QSL10000 | Customer QoS/SLA Management | CHG10000 | Charging | -0.0035
BIM10000 | Billing and Invoice Management | CII10000 | Customer Insight and Intelligence | -0.0030
CEM10000 | Customer Experience Management | CIM10000 | Customer Information Management | -0.0029
BIM10000 | Billing and Invoice Management | BPR10000 | Bill Payments and Receivables Management | 0.0029
CIM10000 | Customer Information Management | CBC10000 | Manage Balances | -0.0028
CII10000 | Customer Insight and Intelligence | CCM10000 | Customer Communication Management | -0.0026

=== chaid_risk_segmentation_report_customer.txt ===

CHAID Risk Segmentation Report
Generated on: 2026-09-20 14:52:21

CHAID Tree Summary:
[[([], {'Critical': 165.0, 'High': 183.0}, (Repetition, p=1.1552379746645288e-77, score=348.0, groups=[['Irregular'], ['Repetitive']]), dof=1)), ([['Irregular'], {'Critical': 165.0, 'High': 0}, <Invalid Chaid Split> - the node only contains single category respondents), ([['Repetitive'], {'Critical': 0, 'High': 183.0}, <Invalid Chaid Split> - the node only contains single category respondents)]

High-Risk Metrics Summary (Critical + High):
Metric ID | Metric Name | Risk Level | Trend | Repetition | Avg Correlation | Inconsistency Level

CSR10101 | Average Response Time | Critical | Decreasing | Irregular | Low | Very High
CSR10102 | First Contact Resolution (FCR) | High | Stable | Repetitive | Low | Very High
CSR10103 | Customer Satisfaction (CSAT) Score | Critical | Decreasing | Irregular | Low | Very High
CSR10104 | Service Level Agreement (SLA) Compliance | Critical | Increasing | Irregular | Low | Very High
CSR10105 | Call Abandonment Rate | Critical | Decreasing | Irregular | Low | Very High
CSR10106 | Agent Productivity | High | Increasing | Repetitive | Low | Very High
CSR10107 | Training Hours per Agent | Critical | Stable | Irregular | Low | Very High
CSR10108 | Knowledge Base Utilization | Critical | Increasing | Irregular | Low | Very High
CSR10109 | Ticket Volume | Critical | Stable | Irregular | Low | Very High
CSR10110 | Customer Retention Rate | Critical | Stable | Irregular | Low | Very High
CSR10111 | Average Handle Time (AHT) | Critical | Increasing | Irregular | Low | Very High
CSR10112 | Escalation Rate | High | Increasing | Repetitive | Low | Very High
CSR10113 | Customer Feedback | Critical | Stable | Irregular | Low | Very High
CSR10114 | Agent Turnover Rate | High | Increasing | Repetitive | Low | Very High
CSR10115 | Technology Utilization | High | Decreasing | Repetitive | Low | Very High

CEM10101 | Customer Effort Score (CES) | High | Decreasing | Repetitive | Low | Very High
CEM10102 | Customer Satisfaction (CSAT) Score | High | Stable | Repetitive | Low | Very High
CEM10103 | Customer Retention Rate | High | Decreasing | Repetitive | Low | Very High
CEM10104 | Customer Churn Rate | High | Increasing | Repetitive | Low | Very High
CEM10105 | Average Handling Time (AHT) | Critical | Increasing | Irregular | Low | Very High
CEM10106 | First Contact Resolution (FCR) | Critical | Decreasing | Irregular | Low | Very High
CEM10107 | Customer Journey Analytics | High | Increasing | Repetitive | Low | Very High
CEM10108 | Service Quality (SQ) Metrics | High | Stable | Repetitive | Low | Very High
CEM10109 | Customer Complaints | Critical | Stable | Irregular | Low | Very High
CEM10110 | Customer Feedback | High | Decreasing | Repetitive | Low | Very High
CEM10111 | Customer Engagement | High | Increasing | Repetitive | Low | Very High
CEM10112 | Customer Lifetime Value (CLV) | High | Stable | Repetitive | Low | Very High
CEM10113 | Customer Segmentation | High | Decreasing | Repetitive | Low | Very High
CEM10114 | Customer Loyalty Programs | Critical | Increasing | Irregular | Low | Very High
ORH10101 | Order Processing Time | High | Stable | Repetitive | Low | Very High
ORH10102 | Order Accuracy Rate | High | Decreasing | Repetitive | Low | Very High
ORH10103 | Order Fulfillment Rate | Critical | Stable | Irregular | Low | Very High
ORH10104 | Order Backlog | High | Decreasing | Repetitive | Low | Very High
ORH10105 | Order Cancellation Rate | High | Decreasing | Repetitive | Low | Very High
ORH10106 | Order Modification Rate | High | Increasing | Repetitive | Low | Very High
ORH10107 | Order Tracking Accuracy | Critical | Increasing | Irregular | Low | Very High
ORH10108 | Customer Order Inquiry Response Time | Critical | Increasing | Irregular | Low | Very High
ORH10109 | Order Completion Rate | Critical | Decreasing | Irregular | Low | Very High
ORH10110 | Order Rejection Rate | High | Stable | Repetitive | Low | Very High
ORH10111 | Order Entry Error Rate | Critical | Decreasing | Irregular | Low | Very High
ORH10112 | Order Status Updates | Critical | Stable | Irregular | Low | Very High
ORH10113 | Order Confirmation Time | High | Decreasing | Repetitive | Low | Very High
ORH10114 | Order Payment Processing Time | High | Increasing | Repetitive | Low | Very High
ORH10115 | Order Returns Rate | Critical | Stable | Irregular | Low | Very High
CRM10101 | Customer Lifetime Value (CLV) | High | Decreasing | Repetitive | Low | Very High
CRM10102 | Customer Retention Rate | Critical | Increasing | Irregular | Low | Very High
CRM10103 | Customer Acquisition Cost (CAC) | Critical | Increasing | Irregular | Low | Very High
CRM10104 | Net Promoter Score (NPS) | High | Stable | Repetitive | Low | Very High
CRM10105 | Customer Satisfaction (CSAT) Score | Critical | Decreasing | Irregular | Low | Very High
CRM10106 | Churn Rate | High | Stable | Repetitive | Low | Very High
CRM10107 | First Contact Resolution (FCR) | High | Stable | Repetitive | Low | Very High
CRM10108 | Average Response Time | High | Stable | Repetitive | Low | Very High
CRM10109 | Customer Engagement Rate | High | Decreasing | Repetitive | Low | Very High
CRM10110 | Sales Conversion Rate | Critical | Decreasing | Irregular | Low | Very High
CRM10111 | Customer Complaints | High | Stable | Repetitive | Low | Very High

CRM10112 | Customer Feedback | Critical | Stable | Irregular | Low | Very High
CRM10113 | Customer Interaction Volume | High | Increasing | Repetitive | Low | Very High
CRM10114 | Referral Rate | Critical | Increasing | Irregular | Low | Very High
CRM10115 | Customer Segmentation | Critical | Stable | Irregular | Low | Very High
CRM10116 | Customer Touchpoint Effectiveness | High | Stable | Repetitive | Low | Very High
CRM10117 | Upsell and Cross-Sell Rates | High | Decreasing | Repetitive | Low | Very High
CRM10118 | Customer Onboarding Time | High | Stable | Repetitive | Low | Very High
CRM10119 | CRM System Utilization | Critical | Decreasing | Irregular | Low | Very High
ITM10101 | First Contact Resolution (FCR) | High | Decreasing | Repetitive | Low | Very High
ITM10102 | Average Handling Time (AHT) | Critical | Stable | Irregular | Low | Very High
ITM10103 | Customer Satisfaction (CSAT) Score | Critical | Decreasing | Irregular | Low | Very High
ITM10104 | Net Promoter Score (NPS) | Critical | Increasing | Irregular | Low | Very High
ITM10105 | Customer Effort Score (CES) | High | Stable | Repetitive | Low | Very High
ITM10106 | Interaction Volume by Channel | High | Increasing | Repetitive | Low | Very High
ITM10107 | Average Response Time | Critical | Stable | Irregular | Low | Very High
ITM10108 | Customer Engagement Rate | High | Stable | Repetitive | Low | Very High
ITM10109 | Interaction Quality Score | Critical | Increasing | Irregular | Low | Very High
ITM10110 | Interaction Abandonment Rate | Critical | Increasing | Irregular | Low | Very High
ITM10111 | Escalation Rate | Critical | Decreasing | Irregular | Low | Very High
ITM10112 | Interaction Conversion Rate | High | Stable | Repetitive | Low | Very High
ITM10113 | Repeat Contact Rate | Critical | Increasing | Irregular | Low | Very High
ITM10114 | Agent Productivity | High | Increasing | Repetitive | Low | Very High
ITM10115 | Interaction Cost | Critical | Stable | Irregular | Low | Very High
ITM10116 | Customer Feedback | Critical | Increasing | Irregular | Low | Very High
ITM10117 | Knowledge Base Utilization | High | Increasing | Repetitive | Low | Very High
ITM10118 | Interaction Channel Preference | High | Increasing | Repetitive | Low | Very High
ITM10119 | Customer Retention Rate | Critical | Stable | Irregular | Low | Very High
CIM10101 | Data Accuracy Rate | Critical | Increasing | Irregular | Low | Very High
CIM10102 | Data Completeness | Critical | Increasing | Irregular | Low | Very High
CIM10103 | Data Timeliness | High | Stable | Repetitive | Low | Very High
CIM10104 | Data Consistency | Critical | Stable | Irregular | Low | Very High
CIM10105 | Data Integrity | High | Decreasing | Repetitive | Low | Very High
CIM10106 | Customer Data Access Frequency | High | Stable | Repetitive | Low | Very High
CIM10107 | Data Security Incidents | Critical | Stable | Irregular | Low | Very High
CIM10108 | Customer Data Update Rate | Critical | Stable | Irregular | Low | Very High
CIM10109 | Data Usage Rate | Critical | Increasing | Irregular | Low | Very High
CIM10110 | Data Redundancy | Critical | Decreasing | Irregular | Low | Very High
CIM10111 | Customer Privacy Compliance | High | Increasing | Repetitive | Low | Very High
CIM10112 | Data Storage Efficiency | High | Decreasing | Repetitive | Low | Very High
CIM10113 | Customer Data Enrichment Rate | Critical | Stable | Irregular | Low | Very High

CIM10114 | Data Matching Accuracy | High | Stable | Repetitive | Low | Very High
CIM10115 | Customer Data Retention Rate | Critical | Increasing | Irregular | Low | Very High
CIM10116 | Data Quality Score | Critical | Stable | Irregular | Low | Very High
CIM10117 | Customer Information Accessibility | Critical | Stable | Irregular | Low | Very High
CIM10118 | Data Anonymization Rate | High | Decreasing | Repetitive | Low | Very High
CIM10119 | Customer Consent Management | Critical | Increasing | Irregular | Low | Very High
CPM10101 | Problem Resolution Time | High | Stable | Repetitive | Low | Very High
CPM10102 | First Contact Resolution (FCR) Rate | Critical | Increasing | Irregular | Low | Very High
CPM10103 | Customer Satisfaction (CSAT) Score | Critical | Stable | Irregular | Low | Very High
CPM10104 | Escalation Rate | High | Decreasing | Repetitive | Low | Very High
CPM10105 | Problem Recurrence Rate | High | Decreasing | Repetitive | Low | Very High
CPM10106 | Resolution Effectiveness | Critical | Increasing | Irregular | Low | Very High
CPM10107 | Average Response Time | Critical | Increasing | Irregular | Low | Very High
CPM10108 | Backlog Volume | High | Stable | Repetitive | Low | Very High
CPM10109 | Root Cause Analysis (RCA) Completion Rate | High | Increasing | Repetitive | Low | Very High
CPM10110 | Customer Retention Rate | Critical | Stable | Irregular | Low | Very High
CPM10111 | Churn Rate | High | Decreasing | Repetitive | Low | Very High
CPM10112 | Customer Complaints Volume | High | Increasing | Repetitive | Low | Very High
CPM10113 | Service Level Agreement (SLA) Compliance | High | Decreasing | Repetitive | Low | Very High
CPM10114 | Cost per Problem Resolution | High | Decreasing | Repetitive | Low | Very High
CPM10115 | Agent Productivity | High | Decreasing | Repetitive | Low | Very High
CPM10116 | Customer Feedback | Critical | Increasing | Irregular | Low | Very High
CPM10117 | Time to Escalate | High | Increasing | Repetitive | Low | Very High
CPM10118 | Resolution Quality Score | Critical | Increasing | Irregular | Low | Very High
CPM10119 | Problem Severity Levels | High | Stable | Repetitive | Low | Very High
QSL10101 | Service Availability | Critical | Stable | Irregular | Low | Very High
QSL10102 | Service Downtime | High | Decreasing | Repetitive | Low | Very High
QSL10103 | Mean Time to Repair (MTTR) | High | Stable | Repetitive | Low | Very High
QSL10104 | Mean Time Between Failures (MTBF) | High | Stable | Repetitive | Low | Very High
QSL10105 | SLA Compliance Rate | High | Stable | Repetitive | Low | Very High
QSL10106 | Incident Response Time | Critical | Increasing | Irregular | Low | Very High
QSL10107 | Incident Resolution Time | High | Stable | Repetitive | Low | Very High
QSL10108 | Customer Satisfaction (CSAT) Score | Critical | Stable | Irregular | Low | Very High
QSL10109 | Net Promoter Score (NPS) | Critical | Stable | Irregular | Low | Very High
QSL10110 | Service Quality Index (SQI) | Critical | Stable | Irregular | Low | Very High
QSL10111 | Uptime Percentage | Critical | Stable | Irregular | Low | Very High
QSL10112 | Service Reliability | High | Stable | Repetitive | Low | Very High
QSL10113 | Network Latency | High | Decreasing | Repetitive | Low | Very High
QSL10114 | Packet Loss Rate | High | Increasing | Repetitive | Low | Very High
QSL10115 | Throughput | High | Decreasing | Repetitive | Low | Very High

QSL10116 | Customer Complaints | Critical | Stable | Irregular | Low | Very High
QSL10117 | Service Degradation Incidents | High | Stable | Repetitive | Low | Very High
QSL10118 | Problem Resolution Time | High | Stable | Repetitive | Low | Very High
QSL10119 | Service Improvement Rate | Critical | Decreasing | Irregular | Low | Very High
BIM10101 | Billing Accuracy Rate | High | Decreasing | Repetitive | Low | Very High
BIM10102 | Invoice Generation Time | Critical | Increasing | Irregular | Low | Very High
BIM10103 | On-Time Invoice Delivery Rate | High | Decreasing | Repetitive | Low | Very High
BIM10104 | Billing Dispute Rate | High | Stable | Repetitive | Low | Very High
BIM10105 | Dispute Resolution Time | Critical | Stable | Irregular | Low | Very High
BIM10106 | Automated Billing Percentage | Critical | Increasing | Irregular | Low | Very High
BIM10107 | Customer Satisfaction (CSAT) with Billing | High | Decreasing | Repetitive | Low | Very High
BIM10108 | Billing Cycle Time | Critical | Decreasing | Irregular | Low | Very High
BIM10109 | Invoice Payment Rate | High | Decreasing | Repetitive | Low | Very High
BIM10110 | Accounts Receivable Turnover | High | Decreasing | Repetitive | Low | Very High
BIM10111 | Days Sales Outstanding (DSO) | Critical | Stable | Irregular | Low | Very High
BIM10112 | Late Payment Rate | Critical | Stable | Irregular | Low | Very High
BIM10113 | Collection Rate | High | Decreasing | Repetitive | Low | Very High
BIM10114 | Credit Note Issuance Rate | Critical | Stable | Irregular | Low | Very High
BIM10115 | Recurring Billing Error Rate | High | Increasing | Repetitive | Low | Very High
BIM10116 | Customer Feedback on Billing | Critical | Increasing | Irregular | Low | Very High
BIM10117 | Billing System Uptime | High | Decreasing | Repetitive | Low | Very High
BIM10118 | Cost per Invoice | Critical | Increasing | Irregular | Low | Very High
BIM10119 | Compliance Rate with Billing Regulations | High | Stable | Repetitive | Low | Very High
BPR10101 | Billing Accuracy Rate | Critical | Increasing | Irregular | Low | Very High
BPR10102 | Invoice Generation Time | Critical | Increasing | Irregular | Low | Very High
BPR10103 | On-Time Invoice Delivery Rate | High | Stable | Repetitive | Low | Very High
BPR10104 | Billing Dispute Rate | High | Stable | Repetitive | Low | Very High
BPR10105 | Dispute Resolution Time | High | Decreasing | Repetitive | Low | Very High
BPR10106 | Automated Billing Percentage | High | Increasing | Repetitive | Low | Very High
BPR10107 | Customer Satisfaction (CSAT) with Billing | High | Increasing | Repetitive | Low | Very High
BPR10108 | Billing Cycle Time | Critical | Increasing | Irregular | Low | Very High
BPR10109 | Invoice Payment Rate | High | Increasing | Repetitive | Low | Very High
BPR10110 | Accounts Receivable Turnover | High | Increasing | Repetitive | Low | Very High
BPR10111 | Days Sales Outstanding (DSO) | High | Decreasing | Repetitive | Low | Very High
BPR10112 | Late Payment Rate | High | Increasing | Repetitive | Low | Very High
BPR10113 | Collection Rate | Critical | Stable | Irregular | Low | Very High
BPR10114 | Credit Note Issuance Rate | High | Stable | Repetitive | Low | Very High
BPR10115 | Recurring Billing Error Rate | Critical | Increasing | Irregular | Low | Very High
BPR10116 | Customer Feedback on Billing | High | Decreasing | Repetitive | Low | Very High
BPR10117 | Billing System Uptime | High | Stable | Repetitive | Low | Very High

BPR10118 | Cost per Invoice | High | Increasing | Repetitive | Low | Very High
BPR10119 | Compliance Rate with Billing Regulations | Critical | Increasing | Irregular | Low | Very High
BIH10101 | Average Response Time | Critical | Decreasing | Irregular | Low | Very High
BIH10102 | First Contact Resolution (FCR) Rate | High | Stable | Repetitive | Low | Very High
BIH10103 | Customer Satisfaction (CSAT) with Billing Inquiries | Critical | Stable | Irregular | Low | Very High
BIH10104 | Inquiry Resolution Time | Critical | Stable | Irregular | Low | Very High
BIH10105 | Billing Inquiry Volume | High | Decreasing | Repetitive | Low | Very High
BIH10106 | Escalation Rate | High | Decreasing | Repetitive | Low | Very High
BIH10107 | Inquiry Accuracy Rate | Critical | Decreasing | Irregular | Low | Very High
BIH10108 | Customer Feedback on Billing Inquiries | Critical | Increasing | Irregular | Low | Very High
BIH10109 | Agent Productivity | High | Decreasing | Repetitive | Low | Very High
BIH10110 | Repeat Inquiry Rate | High | Decreasing | Repetitive | Low | Very High
BIH10111 | Inquiry Abandonment Rate | High | Stable | Repetitive | Low | Very High
BIH10112 | Average Handling Time (AHT) | High | Increasing | Repetitive | Low | Very High
BIH10113 | Billing Inquiry Resolution Effectiveness | Critical | Stable | Irregular | Low | Very High
BIH10114 | Cost per Inquiry | Critical | Stable | Irregular | Low | Very High
BIH10115 | Inquiry Closure Rate | Critical | Stable | Irregular | Low | Very High
BIH10116 | Customer Retention Post-Inquiry | High | Stable | Repetitive | Low | Very High
BIH10117 | Inquiry Channel Utilization | High | Stable | Repetitive | Low | Very High
BIH10118 | Training Hours per Agent | High | Stable | Repetitive | Low | Very High
BIH10119 | Knowledge Base Utilization | High | Increasing | Repetitive | Low | Very High
BEV10101 | Billing Event Accuracy Rate | Critical | Increasing | Irregular | Low | Very High
BEV10102 | Event Processing Time | High | Stable | Repetitive | Low | Very High
BEV10103 | Event Detection Rate | High | Decreasing | Repetitive | Low | Very High
BEV10104 | Event Handling Efficiency | Critical | Stable | Irregular | Low | Very High
BEV10105 | Event Correlation Success Rate | Critical | Decreasing | Irregular | Low | Very High
BEV10106 | Real-Time Event Processing Rate | High | Stable | Repetitive | Low | Very High
BEV10107 | Event Verification Rate | Critical | Stable | Irregular | Low | Very High
BEV10108 | Customer Notification Time | High | Stable | Repetitive | Low | Very High
BEV10109 | Event Dispute Rate | High | Stable | Repetitive | Low | Very High
BEV10110 | Event Resolution Time | High | Stable | Repetitive | Low | Very High
BEV10111 | Event Volume | Critical | Decreasing | Irregular | Low | Very High
BEV10112 | Event Audit Compliance Rate | Critical | Increasing | Irregular | Low | Very High
BEV10113 | Error Correction Rate | Critical | Increasing | Irregular | Low | Very High
BEV10114 | Customer Impact Rate | High | Stable | Repetitive | Low | Very High
BEV10115 | Event-Based Billing Accuracy | High | Increasing | Repetitive | Low | Very High
BEV10116 | Event System Uptime | High | Stable | Repetitive | Low | Very High
BEV10117 | Event Integration Success Rate | Critical | Stable | Irregular | Low | Very High
BEV10118 | Customer Feedback on Billing Events | High | Decreasing | Repetitive | Low | Very High
BEV10119 | Cost per Event Processed | Critical | Increasing | Irregular | Low | Very High

CHG10101 | Charge Accuracy Rate | Critical | Decreasing | Irregular | Low | Very High
CHG10102 | Real-Time Charging Rate | Critical | Stable | Irregular | Low | Very High
CHG10103 | Charging Event Volume | High | Stable | Repetitive | Low | Very High
CHG10104 | Latency in Charging | High | Stable | Repetitive | Low | Very High
CHG10105 | Revenue Assurance Rate | Critical | Decreasing | Irregular | Low | Very High
CHG10106 | Customer Satisfaction (CSAT) with Charging | Critical | Increasing | Irregular | Low | Very High
CHG10107 | Charge Reversal Rate | High | Stable | Repetitive | Low | Very High
CHG10108 | Dispute Resolution Time | Critical | Decreasing | Irregular | Low | Very High
CHG10109 | Charge Dispute Rate | Critical | Decreasing | Irregular | Low | Very High
CHG10110 | Revenue Leakage | Critical | Stable | Irregular | Low | Very High
CHG10111 | Charge Consistency Rate | Critical | Decreasing | Irregular | Low | Very High
CHG10112 | Billing Cycle Time for Charges | High | Decreasing | Repetitive | Low | Very High
CHG10113 | Charge Notification Time | Critical | Stable | Irregular | Low | Very High
CHG10114 | Charging System Uptime | Critical | Decreasing | Irregular | Low | Very High
CHG10115 | Chargeback Rate | High | Stable | Repetitive | Low | Very High
CHG10116 | Rate Plan Accuracy | Critical | Stable | Irregular | Low | Very High
CHG10117 | Automated Charging Percentage | Critical | Increasing | Irregular | Low | Very High
CHG10118 | Cost per Charging Event | High | Stable | Repetitive | Low | Very High
CHG10119 | Compliance Rate with Charging Regulations | Critical | Stable | Irregular | Low | Very High
CBC10101 | Balance Accuracy Rate | High | Increasing | Repetitive | Low | Very High
CBC10102 | Balance Update Frequency | High | Stable | Repetitive | Low | Very High
CBC10103 | Real-Time Balance Availability | High | Decreasing | Repetitive | Low | Very High
CBC10104 | Balance Inquiry Response Time | Critical | Increasing | Irregular | Low | Very High
CBC10105 | Balance Dispute Rate | High | Stable | Repetitive | Low | Very High
CBC10106 | Dispute Resolution Time | High | Increasing | Repetitive | Low | Very High
CBC10107 | Balance Reconciliation Rate | High | Increasing | Repetitive | Low | Very High
CBC10108 | Overdraft Occurrence Rate | Critical | Increasing | Irregular | Low | Very High
CBC10109 | Balance Notification Timeliness | Critical | Decreasing | Irregular | Low | Very High
CBC10110 | Customer Satisfaction (CSAT) with Balance Management | Critical | Decreasing | Irregular | Low | Very High
CBC10111 | Balance Adjustment Rate | High | Increasing | Repetitive | Low | Very High
CBC10112 | Automated Balance Management Rate | High | Stable | Repetitive | Low | Very High
CBC10113 | Cost per Balance Inquiry | High | Decreasing | Repetitive | Low | Very High
CBC10114 | Balance Data Integrity | High | Stable | Repetitive | Low | Very High
CBC10115 | Balance Monitoring Coverage | High | Stable | Repetitive | Low | Very High
CBC10116 | Customer Feedback on Balance Management | High | Decreasing | Repetitive | Low | Very High
CBC10117 | Balance Management System Uptime | High | Decreasing | Repetitive | Low | Very High
CBC10118 | Compliance Rate with Balance Management Regulations | Critical | Decreasing | Irregular | Low | Very High
CBC10119 | Balance Report Accuracy | Critical | Decreasing | Irregular | Low | Very High

RTN10101 | Customer Retention Rate | Critical | Decreasing | Irregular | Low | Very High
RTN10102 | Churn Rate | High | Stable | Repetitive | Low | Very High
RTN10103 | Customer Lifetime Value (CLV) | Critical | Increasing | Irregular | Low | Very High
RTN10104 | Net Promoter Score (NPS) | High | Increasing | Repetitive | Low | Very High
RTN10105 | Customer Satisfaction (CSAT) Score | High | Decreasing | Repetitive | Low | Very High
RTN10106 | Average Customer Tenure | High | Decreasing | Repetitive | Low | Very High
RTN10107 | Repeat Purchase Rate | Critical | Increasing | Irregular | Low | Very High
RTN10108 | Engagement Rate | Critical | Decreasing | Irregular | Low | Very High
RTN10109 | Customer Loyalty Program Participation | Critical | Increasing | Irregular | Low | Very High
RTN10110 | Customer Feedback on Retention Efforts | High | Stable | Repetitive | Low | Very High
RTN10111 | Retention Campaign Effectiveness | Critical | Stable | Irregular | Low | Very High
RTN10112 | Upsell and Cross-Sell Rates | Critical | Increasing | Irregular | Low | Very High
RTN10113 | Customer Support Interaction Frequency | High | Decreasing | Repetitive | Low | Very High
RTN10114 | Service Utilization Rate | Critical | Decreasing | Irregular | Low | Very High
RTN10115 | Customer Exit Survey Results | High | Stable | Repetitive | Low | Very High
RTN10116 | Retention Cost per Customer | High | Stable | Repetitive | Low | Very High
RTN10117 | Product/Service Quality Perception | High | Increasing | Repetitive | Low | Very High
RTN10118 | Customer Complaints Resolution Rate | Critical | Decreasing | Irregular | Low | Very High
RTN10119 | Referral Rate | Critical | Decreasing | Irregular | Low | Very High
CII10101 | Customer Satisfaction (CSAT) Score | High | Decreasing | Repetitive | Low | Very High
CII10102 | Net Promoter Score (NPS) | Critical | Decreasing | Irregular | Low | Very High
CII10103 | Customer Lifetime Value (CLV) | Critical | Decreasing | Irregular | Low | Very High
CII10104 | Churn Rate | High | Stable | Repetitive | Low | Very High
CII10105 | Customer Retention Rate | High | Decreasing | Repetitive | Low | Very High
CII10106 | Customer Segmentation Accuracy | Critical | Stable | Irregular | Low | Very High
CII10107 | Customer Journey Mapping Effectiveness | High | Stable | Repetitive | Low | Very High
CII10108 | Predictive Analytics Accuracy | Critical | Stable | Irregular | Low | Very High
CII10109 | Customer Interaction Volume | Critical | Stable | Irregular | Low | Very High
CII10110 | Customer Feedback Collection Rate | Critical | Stable | Irregular | Low | Very High
CII10111 | Customer Sentiment Analysis | High | Increasing | Repetitive | Low | Very High
CII10112 | Behavioral Analytics Utilization | High | Stable | Repetitive | Low | Very High
CII10113 | Customer Data Integration Rate | Critical | Decreasing | Irregular | Low | Very High
CII10114 | Customer Profile Completeness | Critical | Decreasing | Irregular | Low | Very High
CII10115 | Customer Insight Actionability | High | Decreasing | Repetitive | Low | Very High
CII10116 | Marketing Campaign Effectiveness | High | Stable | Repetitive | Low | Very High
CII10117 | Customer Intelligence Utilization | High | Increasing | Repetitive | Low | Very High
CII10118 | Conversion Rate | Critical | Increasing | Irregular | Low | Very High
CII10119 | Upsell and Cross-Sell Success Rate | Critical | Increasing | Irregular | Low | Very High
CCM10101 | Average Response Time | Critical | Decreasing | Irregular | Low | Very High
CCM10102 | First Contact Resolution (FCR) Rate | Critical | Decreasing | Irregular | Low | Very High

CCM10103 | Customer Satisfaction (CSAT) with Communication | Critical | Stable | Irregular | Low | Very High
CCM10104 | Net Promoter Score (NPS) | High | Decreasing | Repetitive | Low | Very High
CCM10105 | Customer Effort Score (CES) | Critical | Stable | Irregular | Low | Very High
CCM10106 | Communication Channel Effectiveness | Critical | Stable | Irregular | Low | Very High
CCM10107 | Interaction Volume by Channel | High | Increasing | Repetitive | Low | Very High
CCM10108 | Communication Accuracy Rate | High | Increasing | Repetitive | Low | Very High
CCM10109 | Customer Feedback on Communication | Critical | Increasing | Irregular | Low | Very High
CCM10110 | Escalation Rate | Critical | Increasing | Irregular | Low | Very High
CCM10111 | Repeat Contact Rate | High | Increasing | Repetitive | Low | Very High
CCM10112 | Communication Quality Score | High | Stable | Repetitive | Low | Very High
CCM10113 | Customer Engagement Rate | High | Decreasing | Repetitive | Low | Very High
CCM10114 | Average Handling Time (AHT) | High | Decreasing | Repetitive | Low | Very High
CCM10115 | Communication Abandonment Rate | High | Increasing | Repetitive | Low | Very High
CCM10116 | Customer Retention Rate | Critical | Stable | Irregular | Low | Very High
CCM10117 | Training Hours per Agent | Critical | Stable | Irregular | Low | Very High
CCM10118 | Cost per Communication | Critical | Stable | Irregular | Low | Very High
CCM10119 | Knowledge Base Utilization | Critical | Increasing | Irregular | Low | Very High
CSP10101 | Segmentation Accuracy | Critical | Increasing | Irregular | Low | Very High
CSP10102 | Customer Profile Completeness | High | Stable | Repetitive | Low | Very High
CSP10103 | Segmentation Utilization Rate | Critical | Increasing | Irregular | Low | Very High
CSP10104 | Profile Update Frequency | High | Increasing | Repetitive | Low | Very High
CSP10105 | Customer Lifetime Value (CLV) | High | Stable | Repetitive | Low | Very High
CSP10106 | Segment Profitability | High | Increasing | Repetitive | Low | Very High
CSP10107 | Customer Engagement Rate by Segment | High | Decreasing | Repetitive | Low | Very High
CSP10108 | Response Rate to Targeted Campaigns | High | Increasing | Repetitive | Low | Very High
CSP10109 | Conversion Rate by Segment | High | Increasing | Repetitive | Low | Very High
CSP10110 | Customer Retention Rate by Segment | High | Stable | Repetitive | Low | Very High
CSP10111 | Churn Rate by Segment | Critical | Decreasing | Irregular | Low | Very High
CSP10112 | Net Promoter Score (NPS) by Segment | High | Increasing | Repetitive | Low | Very High
CSP10113 | Customer Satisfaction (CSAT) Score by Segment | Critical | Stable | Irregular | Low | Very High
CSP10114 | Segment Size and Growth Rate | Critical | Increasing | Irregular | Low | Very High
CSP10115 | Behavioral Patterns by Segment | Critical | Decreasing | Irregular | Low | Very High
CSP10116 | Demographic Trends | High | Stable | Repetitive | Low | Very High
CSP10117 | Segment-Specific Revenue | High | Stable | Repetitive | Low | Very High
CSP10118 | Customer Feedback by Segment | Critical | Decreasing | Irregular | Low | Very High
CSP10119 | Cost per Segment Profile | Critical | Increasing | Irregular | Low | Very High
CFS10101 | Response Rate | Critical | Increasing | Irregular | Low | Very High
CFS10102 | Customer Satisfaction (CSAT) Score | High | Increasing | Repetitive | Low | Very High
CFS10103 | Net Promoter Score (NPS) | High | Decreasing | Repetitive | Low | Very High
CFS10104 | Feedback Volume | Critical | Stable | Irregular | Low | Very High

CFS10105	Survey Completion Rate	High	Stable	Repetitive	Low	Very High
CFS10106	Feedback Quality Score	High	Stable	Repetitive	Low	Very High
CFS10107	Average Feedback Response Time	Critical	Stable	Irregular	Low	Very High
CFS10108	Sentiment Analysis	High	Stable	Repetitive	Low	Very High
CFS10109	Customer Effort Score (CES)	Critical	Decreasing	Irregular	Low	Very High
CFS10110	Actionable Feedback Rate	High	Stable	Repetitive	Low	Very High
CFS10111	Survey Accuracy	Critical	Decreasing	Irregular	Low	Very High
CFS10112	Follow-up Rate	High	Stable	Repetitive	Low	Very High
CFS10113	Feedback Resolution Time	Critical	Decreasing	Irregular	Low	Very High
CFS10114	Survey Distribution Channels	High	Stable	Repetitive	Low	Very High
CFS10115	Customer Participation Rate	High	Stable	Repetitive	Low	Very High
CFS10116	Feedback Loop Closure Rate	High	Increasing	Repetitive	Low	Very High
CFS10117	Customer Feedback Engagement	High	Stable	Repetitive	Low	Very High
CFS10118	Survey Design Effectiveness	Critical	Decreasing	Irregular	Low	Very High
CFS10119	Cost per Survey Response	Critical	Decreasing	Irregular	Low	Very High

=== inconsistencies_report_customer.txt ===

Inconsistencies Analysis Report
Generated on: 2026-09-20 14:51:41

Metrics Dependent Inconsistencies:
None found.

Metrics Independent Inconsistencies:
60378 pair(s) found; showing top 30 by percentage:

Metric 1 ID	Metric 1 Name	Metric 2 ID	Metric 2 Name	Correlation	Count	Percentage	Over Points
CBC10109	Balance Notification Timeliness	CCM10107	Interaction Volume by Channel	0.06	929	30.95%	3002
CRM10101	Customer Lifetime Value (CLV)	CII10102	Net Promoter Score (NPS)	0.00	919	30.61%	3002
ITM10110	Interaction Abandonment Rate	CCM10104	Net Promoter Score (NPS)	0.02	904	30.11%	3002
CRM10119	CRM System Utilization	RTN10109	Customer Loyalty Program Participation	0.02	900	29.98%	3002
CEM10102	Customer Satisfaction (CSAT) Score	CRM10103	Customer Acquisition Cost (CAC)	0.03	899	29.95%	3002
CEM10105	Average Handling Time (AHT)	BIH10113	Billing Inquiry Resolution Effectiveness	0.05	898	29.91%	3002
CPM10106	Resolution Effectiveness	BPR10101	Billing Accuracy Rate	0.03	898	29.91%	3002
QSL10118	Problem Resolution Time	CFS10113	Feedback Resolution Time	0.05	898	29.91%	3002
CRM10114	Referral Rate	CBC10112	Automated Balance Management Rate	0.02	897	29.88%	3002
BEV10119	Cost per Event Processed	CSP10105	Customer Lifetime Value (CLV)	0.03	896	29.85%	3002
CRM10111	Customer Complaints	BPR10104	Billing Dispute Rate	0.03	895	29.81%	3002

CPM10107 | Average Response Time | BPR10105 | Dispute Resolution Time | 0.03 | 895 | 29.81% | 3002
BIM10104 | Billing Dispute Rate | BIH10117 | Inquiry Channel Utilization | 0.02 | 895 | 29.81% | 3002
RTN10107 | Repeat Purchase Rate | RTN10109 | Customer Loyalty Program Participation | 0.01 | 895 | 29.81% | 3002
CEM10103 | Customer Retention Rate | ITM10108 | Customer Engagement Rate | 0.05 | 893 | 29.75% | 3002
ORH10109 | Order Completion Rate | ITM10111 | Escalation Rate | 0.05 | 892 | 29.71% | 3002
BIH10117 | Inquiry Channel Utilization | BIH10119 | Knowledge Base Utilization | 0.03 | 892 | 29.71% | 3002
CRM10104 | Net Promoter Score (NPS) | BPR10108 | Billing Cycle Time | 0.01 | 891 | 29.68% | 3002
ITM10102 | Average Handling Time (AHT) | CHG10119 | Compliance Rate with Charging Regulations | 0.01 | 891 | 29.68% | 3002
BPR10108 | Billing Cycle Time | CII10105 | Customer Retention Rate | 0.01 | 891 | 29.68% | 3002
BIH10114 | Cost per Inquiry | BEV10102 | Event Processing Time | 0.00 | 891 | 29.68% | 3002
CHG10102 | Real-Time Charging Rate | CBC10103 | Real-Time Balance Availability | 0.03 | 891 | 29.68% | 3002
ORH10105 | Order Cancellation Rate | CII10118 | Conversion Rate | 0.04 | 890 | 29.65% | 3002
CIM10110 | Data Redundancy | BEV10105 | Event Correlation Success Rate | 0.04 | 890 | 29.65% | 3002
CPM10101 | Problem Resolution Time | CSP10110 | Customer Retention Rate by Segment | 0.03 | 890 | 29.65% | 3002
BIH10105 | Billing Inquiry Volume | CBC10103 | Real-Time Balance Availability | 0.04 | 890 | 29.65% | 3002
BEV10108 | Customer Notification Time | CCM10114 | Average Handling Time (AHT) | 0.07 | 890 | 29.65% | 3002
CHG10109 | Charge Dispute Rate | CCM10110 | Escalation Rate | 0.04 | 890 | 29.65% | 3002
CSR10110 | Customer Retention Rate | CHG10108 | Dispute Resolution Time | 0.05 | 889 | 29.61% | 3002
CEM10112 | Customer Lifetime Value (CLV) | CCM10116 | Customer Retention Rate | 0.03 | 889 | 29.61% | 3002

Groups Dependent Inconsistencies:

None found.

Groups Independent Inconsistencies:

171 pair(s) found; showing top 30 by percentage:

Metric 1 ID | Metric 1 Name | Metric 2 ID | Metric 2 Name | Correlation | Count | Percentage | Over Points

CSR10001 | CSR10001 | QSL10001 | QSL10001 | -0.03 | 890 | 29.65% | 3002
CSR10001 | CSR10001 | ITM10001 | ITM10001 | 0.04 | 878 | 29.25% | 3002
ORH10001 | ORH10001 | BEV10001 | BEV10001 | 0.03 | 876 | 29.18% | 3002
ORH10001 | ORH10001 | CFS10001 | CFS10001 | 0.02 | 875 | 29.15% | 3002
CSR10001 | CSR10001 | CCM10001 | CCM10001 | 0.03 | 873 | 29.08% | 3002
ITM10001 | ITM10001 | CSP10001 | CSP10001 | 0.04 | 873 | 29.08% | 3002
ITM10001 | ITM10001 | RTN10001 | RTN10001 | 0.01 | 872 | 29.05% | 3002
CIM10001 | CIM10001 | RTN10001 | RTN10001 | 0.03 | 867 | 28.88% | 3002
BEV10001 | BEV10001 | RTN10001 | RTN10001 | 0.01 | 867 | 28.88% | 3002
CEM10001 | CEM10001 | CSP10001 | CSP10001 | 0.02 | 862 | 28.71% | 3002
CRM10001 | CRM10001 | BPR10001 | BPR10001 | 0.01 | 859 | 28.61% | 3002
ITM10001 | ITM10001 | QSL10001 | QSL10001 | 0.01 | 858 | 28.58% | 3002
CPM10001 | CPM10001 | CII10001 | CII10001 | 0.01 | 858 | 28.58% | 3002

CII10001 | CII10001 | CCM10001 | CCM10001 | -0.00 | 857 | 28.55% | 3002
ORH10001 | ORH10001 | CCM10001 | CCM10001 | 0.01 | 854 | 28.45% | 3002
ORH10001 | ORH10001 | ITM10001 | ITM10001 | 0.02 | 853 | 28.41% | 3002
ITM10001 | ITM10001 | CCM10001 | CCM10001 | 0.01 | 853 | 28.41% | 3002
QSL10001 | QSL10001 | RTN10001 | RTN10001 | 0.00 | 853 | 28.41% | 3002
BPR10001 | BPR10001 | BIH10001 | BIH10001 | -0.02 | 852 | 28.38% | 3002
BIH10001 | BIH10001 | CSP10001 | CSP10001 | -0.00 | 852 | 28.38% | 3002
CHG10001 | CHG10001 | CFS10001 | CFS10001 | -0.02 | 852 | 28.38% | 3002
CEM10001 | CEM10001 | CRM10001 | CRM10001 | -0.02 | 848 | 28.25% | 3002
ITM10001 | ITM10001 | BEV10001 | BEV10001 | -0.00 | 848 | 28.25% | 3002
CSP10001 | CSP10001 | CFS10001 | CFS10001 | -0.01 | 846 | 28.18% | 3002
ITM10001 | ITM10001 | CPM10001 | CPM10001 | -0.01 | 845 | 28.15% | 3002
QSL10001 | QSL10001 | BIH10001 | BIH10001 | 0.01 | 845 | 28.15% | 3002
BIM10001 | BIM10001 | BIH10001 | BIH10001 | 0.03 | 845 | 28.15% | 3002
ORH10001 | ORH10001 | CBC10001 | CBC10001 | -0.01 | 844 | 28.11% | 3002
CRM10001 | CRM10001 | BEV10001 | BEV10001 | -0.01 | 844 | 28.11% | 3002
ITM10001 | ITM10001 | CII10001 | CII10001 | -0.00 | 843 | 28.08% | 3002

Sub_Modules Dependent Inconsistencies:

None found.

Sub_Modules Independent Inconsistencies:

171 pair(s) found; showing top 30 by percentage:

Metric 1 ID | Metric 1 Name | Metric 2 ID | Metric 2 Name | Correlation | Count | Percentage | Over Points

CSR10000 | CSR10000 | QSL10000 | QSL10000 | -0.03 | 890 | 29.65% | 3002
CSR10000 | CSR10000 | ITM10000 | ITM10000 | 0.04 | 878 | 29.25% | 3002
ORH10000 | ORH10000 | BEV10000 | BEV10000 | 0.03 | 876 | 29.18% | 3002
ORH10000 | ORH10000 | CFS10000 | CFS10000 | 0.02 | 875 | 29.15% | 3002
CSR10000 | CSR10000 | CCM10000 | CCM10000 | 0.03 | 873 | 29.08% | 3002
ITM10000 | ITM10000 | CSP10000 | CSP10000 | 0.04 | 873 | 29.08% | 3002
ITM10000 | ITM10000 | RTN10000 | RTN10000 | 0.01 | 872 | 29.05% | 3002
CIM10000 | CIM10000 | RTN10000 | RTN10000 | 0.03 | 867 | 28.88% | 3002
BEV10000 | BEV10000 | RTN10000 | RTN10000 | 0.01 | 867 | 28.88% | 3002
CEM10000 | CEM10000 | CSP10000 | CSP10000 | 0.02 | 862 | 28.71% | 3002
CRM10000 | CRM10000 | BPR10000 | BPR10000 | 0.01 | 859 | 28.61% | 3002
ITM10000 | ITM10000 | QSL10000 | QSL10000 | 0.01 | 858 | 28.58% | 3002
CPM10000 | CPM10000 | CII10000 | CII10000 | 0.01 | 858 | 28.58% | 3002
CII10000 | CII10000 | CCM10000 | CCM10000 | -0.00 | 857 | 28.55% | 3002
ORH10000 | ORH10000 | CCM10000 | CCM10000 | 0.01 | 854 | 28.45% | 3002
ORH10000 | ORH10000 | ITM10000 | ITM10000 | 0.02 | 853 | 28.41% | 3002
ITM10000 | ITM10000 | CCM10000 | CCM10000 | 0.01 | 853 | 28.41% | 3002

QSL10000 | QSL10000 | RTN10000 | RTN10000 | 0.00 | 853 | 28.41% | 3002
BPR10000 | BPR10000 | BIH10000 | BIH10000 | -0.02 | 852 | 28.38% | 3002
BIH10000 | BIH10000 | CSP10000 | CSP10000 | -0.00 | 852 | 28.38% | 3002
CHG10000 | CHG10000 | CFS10000 | CFS10000 | -0.02 | 852 | 28.38% | 3002
CEM10000 | CEM10000 | CRM10000 | CRM10000 | -0.02 | 848 | 28.25% | 3002
ITM10000 | ITM10000 | BEV10000 | BEV10000 | -0.00 | 848 | 28.25% | 3002
CSP10000 | CSP10000 | CFS10000 | CFS10000 | -0.01 | 846 | 28.18% | 3002
ITM10000 | ITM10000 | CPM10000 | CPM10000 | -0.01 | 845 | 28.15% | 3002
QSL10000 | QSL10000 | BIH10000 | BIH10000 | 0.01 | 845 | 28.15% | 3002
BIM10000 | BIM10000 | BIH10000 | BIH10000 | 0.03 | 845 | 28.15% | 3002
ORH10000 | ORH10000 | CBC10000 | CBC10000 | -0.01 | 844 | 28.11% | 3002
CRM10000 | CRM10000 | BEV10000 | BEV10000 | -0.01 | 844 | 28.11% | 3002
ITM10000 | ITM10000 | CII10000 | CII10000 | -0.00 | 843 | 28.08% | 3002

=== trends_and_repetitions_report_customer.txt ===

Metrics Trends:

CSR10101: Decreasing
CSR10102: Stable
CSR10103: Decreasing
CSR10104: Increasing
CSR10105: Decreasing
CSR10106: Increasing
CSR10107: Stable
CSR10108: Increasing
CSR10109: Stable
CSR10110: Stable
CSR10111: Increasing
CSR10112: Increasing
CSR10113: Stable
CSR10114: Increasing
CSR10115: Decreasing
CEM10101: Decreasing
CEM10102: Stable
CEM10103: Decreasing
CEM10104: Increasing
CEM10105: Increasing
CEM10106: Decreasing
CEM10107: Increasing
CEM10108: Stable

CEM10109: Stable
CEM10110: Decreasing
CEM10111: Increasing
CEM10112: Stable
CEM10113: Decreasing
CEM10114: Increasing
ORH10101: Stable
ORH10102: Decreasing
ORH10103: Stable
ORH10104: Decreasing
ORH10105: Decreasing
ORH10106: Increasing
ORH10107: Increasing
ORH10108: Increasing
ORH10109: Decreasing
ORH10110: Stable
ORH10111: Decreasing
ORH10112: Stable
ORH10113: Decreasing
ORH10114: Increasing
ORH10115: Stable
CRM10101: Decreasing
CRM10102: Increasing
CRM10103: Increasing
CRM10104: Stable
CRM10105: Decreasing
CRM10106: Stable
CRM10107: Stable
CRM10108: Stable
CRM10109: Decreasing
CRM10110: Decreasing
CRM10111: Stable
CRM10112: Stable
CRM10113: Increasing
CRM10114: Increasing
CRM10115: Stable
CRM10116: Stable
CRM10117: Decreasing
CRM10118: Stable
CRM10119: Decreasing

ITM10101: Decreasing
ITM10102: Stable
ITM10103: Decreasing
ITM10104: Increasing
ITM10105: Stable
ITM10106: Increasing
ITM10107: Stable
ITM10108: Stable
ITM10109: Increasing
ITM10110: Increasing
ITM10111: Decreasing
ITM10112: Stable
ITM10113: Increasing
ITM10114: Increasing
ITM10115: Stable
ITM10116: Increasing
ITM10117: Increasing
ITM10118: Increasing
ITM10119: Stable
CIM10101: Increasing
CIM10102: Increasing
CIM10103: Stable
CIM10104: Stable
CIM10105: Decreasing
CIM10106: Stable
CIM10107: Stable
CIM10108: Stable
CIM10109: Increasing
CIM10110: Decreasing
CIM10111: Increasing
CIM10112: Decreasing
CIM10113: Stable
CIM10114: Stable
CIM10115: Increasing
CIM10116: Stable
CIM10117: Stable
CIM10118: Decreasing
CIM10119: Increasing
CPM10101: Stable
CPM10102: Increasing

CPM10103: Stable
CPM10104: Decreasing
CPM10105: Decreasing
CPM10106: Increasing
CPM10107: Increasing
CPM10108: Stable
CPM10109: Increasing
CPM10110: Stable
CPM10111: Decreasing
CPM10112: Increasing
CPM10113: Decreasing
CPM10114: Decreasing
CPM10115: Decreasing
CPM10116: Increasing
CPM10117: Increasing
CPM10118: Increasing
CPM10119: Stable
QSL10101: Stable
QSL10102: Decreasing
QSL10103: Stable
QSL10104: Stable
QSL10105: Stable
QSL10106: Increasing
QSL10107: Stable
QSL10108: Stable
QSL10109: Stable
QSL10110: Stable
QSL10111: Stable
QSL10112: Stable
QSL10113: Decreasing
QSL10114: Increasing
QSL10115: Decreasing
QSL10116: Stable
QSL10117: Stable
QSL10118: Stable
QSL10119: Decreasing
BIM10101: Decreasing
BIM10102: Increasing
BIM10103: Decreasing
BIM10104: Stable

BIM10105: Stable
BIM10106: Increasing
BIM10107: Decreasing
BIM10108: Decreasing
BIM10109: Decreasing
BIM10110: Decreasing
BIM10111: Stable
BIM10112: Stable
BIM10113: Decreasing
BIM10114: Stable
BIM10115: Increasing
BIM10116: Increasing
BIM10117: Decreasing
BIM10118: Increasing
BIM10119: Stable
BPR10101: Increasing
BPR10102: Increasing
BPR10103: Stable
BPR10104: Stable
BPR10105: Decreasing
BPR10106: Increasing
BPR10107: Increasing
BPR10108: Increasing
BPR10109: Increasing
BPR10110: Increasing
BPR10111: Decreasing
BPR10112: Increasing
BPR10113: Stable
BPR10114: Stable
BPR10115: Increasing
BPR10116: Decreasing
BPR10117: Stable
BPR10118: Increasing
BPR10119: Increasing
BIH10101: Decreasing
BIH10102: Stable
BIH10103: Stable
BIH10104: Stable
BIH10105: Decreasing
BIH10106: Decreasing

BIH10107: Decreasing
BIH10108: Increasing
BIH10109: Decreasing
BIH10110: Decreasing
BIH10111: Stable
BIH10112: Increasing
BIH10113: Stable
BIH10114: Stable
BIH10115: Stable
BIH10116: Stable
BIH10117: Stable
BIH10118: Stable
BIH10119: Increasing
BEV10101: Increasing
BEV10102: Stable
BEV10103: Decreasing
BEV10104: Stable
BEV10105: Decreasing
BEV10106: Stable
BEV10107: Stable
BEV10108: Stable
BEV10109: Stable
BEV10110: Stable
BEV10111: Decreasing
BEV10112: Increasing
BEV10113: Increasing
BEV10114: Stable
BEV10115: Increasing
BEV10116: Stable
BEV10117: Stable
BEV10118: Decreasing
BEV10119: Increasing
CHG10101: Decreasing
CHG10102: Stable
CHG10103: Stable
CHG10104: Stable
CHG10105: Decreasing
CHG10106: Increasing
CHG10107: Stable
CHG10108: Decreasing

CHG10109: Decreasing
CHG10110: Stable
CHG10111: Decreasing
CHG10112: Decreasing
CHG10113: Stable
CHG10114: Decreasing
CHG10115: Stable
CHG10116: Stable
CHG10117: Increasing
CHG10118: Stable
CHG10119: Stable
CBC10101: Increasing
CBC10102: Stable
CBC10103: Decreasing
CBC10104: Increasing
CBC10105: Stable
CBC10106: Increasing
CBC10107: Increasing
CBC10108: Increasing
CBC10109: Decreasing
CBC10110: Decreasing
CBC10111: Increasing
CBC10112: Stable
CBC10113: Decreasing
CBC10114: Stable
CBC10115: Stable
CBC10116: Decreasing
CBC10117: Decreasing
CBC10118: Decreasing
CBC10119: Decreasing
RTN10101: Decreasing
RTN10102: Stable
RTN10103: Increasing
RTN10104: Increasing
RTN10105: Decreasing
RTN10106: Decreasing
RTN10107: Increasing
RTN10108: Decreasing
RTN10109: Increasing
RTN10110: Stable

RTN10111: Stable
RTN10112: Increasing
RTN10113: Decreasing
RTN10114: Decreasing
RTN10115: Stable
RTN10116: Stable
RTN10117: Increasing
RTN10118: Decreasing
RTN10119: Decreasing
CII10101: Decreasing
CII10102: Decreasing
CII10103: Decreasing
CII10104: Stable
CII10105: Decreasing
CII10106: Stable
CII10107: Stable
CII10108: Stable
CII10109: Stable
CII10110: Stable
CII10111: Increasing
CII10112: Stable
CII10113: Decreasing
CII10114: Decreasing
CII10115: Decreasing
CII10116: Stable
CII10117: Increasing
CII10118: Increasing
CII10119: Increasing
CCM10101: Decreasing
CCM10102: Decreasing
CCM10103: Stable
CCM10104: Decreasing
CCM10105: Stable
CCM10106: Stable
CCM10107: Increasing
CCM10108: Increasing
CCM10109: Increasing
CCM10110: Increasing
CCM10111: Increasing
CCM10112: Stable

CCM10113: Decreasing
CCM10114: Decreasing
CCM10115: Increasing
CCM10116: Stable
CCM10117: Stable
CCM10118: Stable
CCM10119: Increasing
CSP10101: Increasing
CSP10102: Stable
CSP10103: Increasing
CSP10104: Increasing
CSP10105: Stable
CSP10106: Increasing
CSP10107: Decreasing
CSP10108: Increasing
CSP10109: Increasing
CSP10110: Stable
CSP10111: Decreasing
CSP10112: Increasing
CSP10113: Stable
CSP10114: Increasing
CSP10115: Decreasing
CSP10116: Stable
CSP10117: Stable
CSP10118: Decreasing
CSP10119: Increasing
CFS10101: Increasing
CFS10102: Increasing
CFS10103: Decreasing
CFS10104: Stable
CFS10105: Stable
CFS10106: Stable
CFS10107: Stable
CFS10108: Stable
CFS10109: Decreasing
CFS10110: Stable
CFS10111: Decreasing
CFS10112: Stable
CFS10113: Decreasing
CFS10114: Stable

CFS10115: Stable

CFS10116: Increasing

CFS10117: Stable

CFS10118: Decreasing

CFS10119: Decreasing

Metrics Repetitions (Fourier Analysis):

CSR10101: No clear repetition (potential instability)

CSR10102: Dominant period: 2.04 units (repetitive)

CSR10103: No clear repetition (potential instability)

CSR10104: No clear repetition (potential instability)

CSR10105: No clear repetition (potential instability)

CSR10106: Dominant period: 2.13 units (repetitive)

CSR10107: No clear repetition (potential instability)

CSR10108: No clear repetition (potential instability)

CSR10109: No clear repetition (potential instability)

CSR10110: No clear repetition (potential instability)

CSR10111: No clear repetition (potential instability)

CSR10112: Dominant period: 9.15 units (repetitive)

CSR10113: No clear repetition (potential instability)

CSR10114: Dominant period: 13.22 units (repetitive)

CSR10115: Dominant period: 3.70 units (repetitive)

CEM10101: Dominant period: 5.17 units (repetitive)

CEM10102: Dominant period: 272.91 units (repetitive)

CEM10103: Dominant period: 3.72 units (repetitive)

CEM10104: Dominant period: 2.03 units (repetitive)

CEM10105: No clear repetition (potential instability)

CEM10106: No clear repetition (potential instability)

CEM10107: Dominant period: 18.19 units (repetitive)

CEM10108: Dominant period: 4.15 units (repetitive)

CEM10109: No clear repetition (potential instability)

CEM10110: Dominant period: 13.71 units (repetitive)

CEM10111: Dominant period: 5.06 units (repetitive)

CEM10112: Dominant period: 44.81 units (repetitive)

CEM10113: Dominant period: 5.73 units (repetitive)

CEM10114: No clear repetition (potential instability)

ORH10101: Dominant period: 2.14 units (repetitive)

ORH10102: Dominant period: 8.18 units (repetitive)

ORH10103: No clear repetition (potential instability)

ORH10104: Dominant period: 2.19 units (repetitive)

ORH10105: Dominant period: 2.43 units (repetitive)
ORH10106: Dominant period: 6.11 units (repetitive)
ORH10107: No clear repetition (potential instability)
ORH10108: No clear repetition (potential instability)
ORH10109: No clear repetition (potential instability)
ORH10110: Dominant period: 2.14 units (repetitive)
ORH10111: No clear repetition (potential instability)
ORH10112: No clear repetition (potential instability)
ORH10113: Dominant period: 2.91 units (repetitive)
ORH10114: Dominant period: 2.07 units (repetitive)
ORH10115: No clear repetition (potential instability)
CRM10101: Dominant period: 2.47 units (repetitive)
CRM10102: No clear repetition (potential instability)
CRM10103: No clear repetition (potential instability)
CRM10104: Dominant period: 9.56 units (repetitive)
CRM10105: No clear repetition (potential instability)
CRM10106: Dominant period: 9.88 units (repetitive)
CRM10107: Dominant period: 8.53 units (repetitive)
CRM10108: Dominant period: 5.78 units (repetitive)
CRM10109: Dominant period: 13.05 units (repetitive)
CRM10110: No clear repetition (potential instability)
CRM10111: Dominant period: 13.52 units (repetitive)
CRM10112: No clear repetition (potential instability)
CRM10113: Dominant period: 52.67 units (repetitive)
CRM10114: No clear repetition (potential instability)
CRM10115: No clear repetition (potential instability)
CRM10116: Dominant period: 158.00 units (repetitive)
CRM10117: Dominant period: 16.05 units (repetitive)
CRM10118: Dominant period: 2.63 units (repetitive)
CRM10119: No clear repetition (potential instability)
ITM10101: Dominant period: 5.04 units (repetitive)
ITM10102: No clear repetition (potential instability)
ITM10103: No clear repetition (potential instability)
ITM10104: No clear repetition (potential instability)
ITM10105: Dominant period: 24.81 units (repetitive)
ITM10106: Dominant period: 3.43 units (repetitive)
ITM10107: No clear repetition (potential instability)
ITM10108: Dominant period: 2.67 units (repetitive)
ITM10109: No clear repetition (potential instability)
ITM10110: No clear repetition (potential instability)

ITM10111: No clear repetition (potential instability)
ITM10112: Dominant period: 3.44 units (repetitive)
ITM10113: No clear repetition (potential instability)
ITM10114: Dominant period: 46.91 units (repetitive)
ITM10115: No clear repetition (potential instability)
ITM10116: No clear repetition (potential instability)
ITM10117: Dominant period: 4.36 units (repetitive)
ITM10118: Dominant period: 2.70 units (repetitive)
ITM10119: No clear repetition (potential instability)
CIM10101: No clear repetition (potential instability)
CIM10102: No clear repetition (potential instability)
CIM10103: Dominant period: 2.53 units (repetitive)
CIM10104: No clear repetition (potential instability)
CIM10105: Dominant period: 3.34 units (repetitive)
CIM10106: Dominant period: 5.89 units (repetitive)
CIM10107: No clear repetition (potential instability)
CIM10108: No clear repetition (potential instability)
CIM10109: No clear repetition (potential instability)
CIM10110: No clear repetition (potential instability)
CIM10111: Dominant period: 6.39 units (repetitive)
CIM10112: Dominant period: 13.58 units (repetitive)
CIM10113: No clear repetition (potential instability)
CIM10114: Dominant period: 61.27 units (repetitive)
CIM10115: No clear repetition (potential instability)
CIM10116: No clear repetition (potential instability)
CIM10117: No clear repetition (potential instability)
CIM10118: Dominant period: 3.53 units (repetitive)
CIM10119: No clear repetition (potential instability)
CPM10101: Dominant period: 3.22 units (repetitive)
CPM10102: No clear repetition (potential instability)
CPM10103: No clear repetition (potential instability)
CPM10104: Dominant period: 3.99 units (repetitive)
CPM10105: Dominant period: 9.50 units (repetitive)
CPM10106: No clear repetition (potential instability)
CPM10107: No clear repetition (potential instability)
CPM10108: Dominant period: 4.80 units (repetitive)
CPM10109: Dominant period: 15.80 units (repetitive)
CPM10110: No clear repetition (potential instability)
CPM10111: Dominant period: 2.04 units (repetitive)
CPM10112: Dominant period: 4.45 units (repetitive)

CPM10113: Dominant period: 2.27 units (repetitive)
CPM10114: Dominant period: 6.55 units (repetitive)
CPM10115: Dominant period: 3.56 units (repetitive)
CPM10116: No clear repetition (potential instability)
CPM10117: Dominant period: 7.52 units (repetitive)
CPM10118: No clear repetition (potential instability)
CPM10119: Dominant period: 3.29 units (repetitive)
QSL10101: No clear repetition (potential instability)
QSL10102: Dominant period: 4.19 units (repetitive)
QSL10103: Dominant period: 19.00 units (repetitive)
QSL10104: Dominant period: 9.56 units (repetitive)
QSL10105: Dominant period: 11.00 units (repetitive)
QSL10106: No clear repetition (potential instability)
QSL10107: Dominant period: 6.28 units (repetitive)
QSL10108: No clear repetition (potential instability)
QSL10109: No clear repetition (potential instability)
QSL10110: No clear repetition (potential instability)
QSL10111: No clear repetition (potential instability)
QSL10112: Dominant period: 4.44 units (repetitive)
QSL10113: Dominant period: 4.15 units (repetitive)
QSL10114: Dominant period: 12.10 units (repetitive)
QSL10115: Dominant period: 3.19 units (repetitive)
QSL10116: No clear repetition (potential instability)
QSL10117: Dominant period: 272.91 units (repetitive)
QSL10118: Dominant period: 2.25 units (repetitive)
QSL10119: No clear repetition (potential instability)
BIM10101: Dominant period: 14.03 units (repetitive)
BIM10102: No clear repetition (potential instability)
BIM10103: Dominant period: 5.33 units (repetitive)
BIM10104: Dominant period: 6.90 units (repetitive)
BIM10105: No clear repetition (potential instability)
BIM10106: No clear repetition (potential instability)
BIM10107: Dominant period: 187.62 units (repetitive)
BIM10108: No clear repetition (potential instability)
BIM10109: Dominant period: 8.27 units (repetitive)
BIM10110: Dominant period: 4.87 units (repetitive)
BIM10111: No clear repetition (potential instability)
BIM10112: No clear repetition (potential instability)
BIM10113: Dominant period: 2.63 units (repetitive)
BIM10114: No clear repetition (potential instability)

BIM10115: Dominant period: 2.35 units (repetitive)
BIM10116: No clear repetition (potential instability)
BIM10117: Dominant period: 4.32 units (repetitive)
BIM10118: No clear repetition (potential instability)
BIM10119: Dominant period: 2.43 units (repetitive)
BPR10101: No clear repetition (potential instability)
BPR10102: No clear repetition (potential instability)
BPR10103: Dominant period: 24.81 units (repetitive)
BPR10104: Dominant period: 3.34 units (repetitive)
BPR10105: Dominant period: 2.04 units (repetitive)
BPR10106: Dominant period: 2.01 units (repetitive)
BPR10107: Dominant period: 176.59 units (repetitive)
BPR10108: No clear repetition (potential instability)
BPR10109: Dominant period: 4.00 units (repetitive)
BPR10110: Dominant period: 26.80 units (repetitive)
BPR10111: Dominant period: 31.27 units (repetitive)
BPR10112: Dominant period: 2.71 units (repetitive)
BPR10113: No clear repetition (potential instability)
BPR10114: Dominant period: 8.11 units (repetitive)
BPR10115: No clear repetition (potential instability)
BPR10116: Dominant period: 5.06 units (repetitive)
BPR10117: Dominant period: 6.79 units (repetitive)
BPR10118: Dominant period: 6.36 units (repetitive)
BPR10119: No clear repetition (potential instability)
BIH10101: No clear repetition (potential instability)
BIH10102: Dominant period: 24.02 units (repetitive)
BIH10103: No clear repetition (potential instability)
BIH10104: No clear repetition (potential instability)
BIH10105: Dominant period: 14.23 units (repetitive)
BIH10106: Dominant period: 2.55 units (repetitive)
BIH10107: No clear repetition (potential instability)
BIH10108: No clear repetition (potential instability)
BIH10109: Dominant period: 3.36 units (repetitive)
BIH10110: Dominant period: 2.98 units (repetitive)
BIH10111: Dominant period: 5.05 units (repetitive)
BIH10112: Dominant period: 2.24 units (repetitive)
BIH10113: No clear repetition (potential instability)
BIH10114: No clear repetition (potential instability)
BIH10115: No clear repetition (potential instability)
BIH10116: Dominant period: 2.82 units (repetitive)

BIH10117: Dominant period: 19.88 units (repetitive)
BIH10118: Dominant period: 2.54 units (repetitive)
BIH10119: Dominant period: 6.11 units (repetitive)
BEV10101: No clear repetition (potential instability)
BEV10102: Dominant period: 3.86 units (repetitive)
BEV10103: Dominant period: 16.14 units (repetitive)
BEV10104: No clear repetition (potential instability)
BEV10105: No clear repetition (potential instability)
BEV10106: Dominant period: 272.91 units (repetitive)
BEV10107: No clear repetition (potential instability)
BEV10108: Dominant period: 2.35 units (repetitive)
BEV10109: Dominant period: 11.08 units (repetitive)
BEV10110: Dominant period: 46.91 units (repetitive)
BEV10111: No clear repetition (potential instability)
BEV10112: No clear repetition (potential instability)
BEV10113: No clear repetition (potential instability)
BEV10114: Dominant period: 428.86 units (repetitive)
BEV10115: Dominant period: 24.41 units (repetitive)
BEV10116: Dominant period: 3.30 units (repetitive)
BEV10117: No clear repetition (potential instability)
BEV10118: Dominant period: 2.22 units (repetitive)
BEV10119: No clear repetition (potential instability)
CHG10101: No clear repetition (potential instability)
CHG10102: No clear repetition (potential instability)
CHG10103: Dominant period: 2.14 units (repetitive)
CHG10104: Dominant period: 93.81 units (repetitive)
CHG10105: No clear repetition (potential instability)
CHG10106: No clear repetition (potential instability)
CHG10107: Dominant period: 2.55 units (repetitive)
CHG10108: No clear repetition (potential instability)
CHG10109: No clear repetition (potential instability)
CHG10110: No clear repetition (potential instability)
CHG10111: No clear repetition (potential instability)
CHG10112: Dominant period: 2.31 units (repetitive)
CHG10113: No clear repetition (potential instability)
CHG10114: No clear repetition (potential instability)
CHG10115: Dominant period: 9.56 units (repetitive)
CHG10116: No clear repetition (potential instability)
CHG10117: No clear repetition (potential instability)
CHG10118: Dominant period: 5.70 units (repetitive)

CHG10119: No clear repetition (potential instability)
CBC10101: Dominant period: 44.81 units (repetitive)
CBC10102: Dominant period: 2.51 units (repetitive)
CBC10103: Dominant period: 3.91 units (repetitive)
CBC10104: No clear repetition (potential instability)
CBC10105: Dominant period: 6.61 units (repetitive)
CBC10106: Dominant period: 6.78 units (repetitive)
CBC10107: Dominant period: 6.10 units (repetitive)
CBC10108: No clear repetition (potential instability)
CBC10109: No clear repetition (potential instability)
CBC10110: No clear repetition (potential instability)
CBC10111: Dominant period: 9.59 units (repetitive)
CBC10112: Dominant period: 3.47 units (repetitive)
CBC10113: Dominant period: 12.35 units (repetitive)
CBC10114: Dominant period: 4.36 units (repetitive)
CBC10115: Dominant period: 3.67 units (repetitive)
CBC10116: Dominant period: 3.32 units (repetitive)
CBC10117: Dominant period: 2.72 units (repetitive)
CBC10118: No clear repetition (potential instability)
CBC10119: No clear repetition (potential instability)
RTN10101: No clear repetition (potential instability)
RTN10102: Dominant period: 90.97 units (repetitive)
RTN10103: No clear repetition (potential instability)
RTN10104: Dominant period: 5.29 units (repetitive)
RTN10105: Dominant period: 16.40 units (repetitive)
RTN10106: Dominant period: 4.98 units (repetitive)
RTN10107: No clear repetition (potential instability)
RTN10108: No clear repetition (potential instability)
RTN10109: No clear repetition (potential instability)
RTN10110: Dominant period: 158.00 units (repetitive)
RTN10111: No clear repetition (potential instability)
RTN10112: No clear repetition (potential instability)
RTN10113: Dominant period: 21.60 units (repetitive)
RTN10114: No clear repetition (potential instability)
RTN10115: Dominant period: 4.34 units (repetitive)
RTN10116: Dominant period: 37.52 units (repetitive)
RTN10117: Dominant period: 5.27 units (repetitive)
RTN10118: No clear repetition (potential instability)
RTN10119: No clear repetition (potential instability)
CII10101: Dominant period: 7.16 units (repetitive)

CII10102: No clear repetition (potential instability)
CII10103: No clear repetition (potential instability)
CII10104: Dominant period: 2.91 units (repetitive)
CII10105: Dominant period: 38.00 units (repetitive)
CII10106: No clear repetition (potential instability)
CII10107: Dominant period: 10.42 units (repetitive)
CII10108: No clear repetition (potential instability)
CII10109: No clear repetition (potential instability)
CII10110: No clear repetition (potential instability)
CII10111: Dominant period: 3.33 units (repetitive)
CII10112: Dominant period: 12.40 units (repetitive)
CII10113: No clear repetition (potential instability)
CII10114: No clear repetition (potential instability)
CII10115: Dominant period: 3.24 units (repetitive)
CII10116: Dominant period: 3.06 units (repetitive)
CII10117: Dominant period: 11.82 units (repetitive)
CII10118: No clear repetition (potential instability)
CII10119: No clear repetition (potential instability)
CCM10101: No clear repetition (potential instability)
CCM10102: No clear repetition (potential instability)
CCM10103: No clear repetition (potential instability)
CCM10104: Dominant period: 2.48 units (repetitive)
CCM10105: No clear repetition (potential instability)
CCM10106: No clear repetition (potential instability)
CCM10107: Dominant period: 3.52 units (repetitive)
CCM10108: Dominant period: 5.63 units (repetitive)
CCM10109: No clear repetition (potential instability)
CCM10110: No clear repetition (potential instability)
CCM10111: Dominant period: 6.58 units (repetitive)
CCM10112: Dominant period: 3.02 units (repetitive)
CCM10113: Dominant period: 10.72 units (repetitive)
CCM10114: Dominant period: 4.00 units (repetitive)
CCM10115: Dominant period: 9.62 units (repetitive)
CCM10116: No clear repetition (potential instability)
CCM10117: No clear repetition (potential instability)
CCM10118: No clear repetition (potential instability)
CCM10119: No clear repetition (potential instability)
CSP10101: No clear repetition (potential instability)
CSP10102: Dominant period: 7.74 units (repetitive)
CSP10103: No clear repetition (potential instability)

CSP10104: Dominant period: 2.36 units (repetitive)
CSP10105: Dominant period: 2.02 units (repetitive)
CSP10106: Dominant period: 2.41 units (repetitive)
CSP10107: Dominant period: 5.24 units (repetitive)
CSP10108: Dominant period: 5.87 units (repetitive)
CSP10109: Dominant period: 25.88 units (repetitive)
CSP10110: Dominant period: 4.44 units (repetitive)
CSP10111: No clear repetition (potential instability)
CSP10112: Dominant period: 7.32 units (repetitive)
CSP10113: No clear repetition (potential instability)
CSP10114: No clear repetition (potential instability)
CSP10115: No clear repetition (potential instability)
CSP10116: Dominant period: 8.73 units (repetitive)
CSP10117: Dominant period: 10.80 units (repetitive)
CSP10118: No clear repetition (potential instability)
CSP10119: No clear repetition (potential instability)
CFS10101: No clear repetition (potential instability)
CFS10102: Dominant period: 2.19 units (repetitive)
CFS10103: Dominant period: 3.52 units (repetitive)
CFS10104: No clear repetition (potential instability)
CFS10105: Dominant period: 3.09 units (repetitive)
CFS10106: Dominant period: 5.21 units (repetitive)
CFS10107: No clear repetition (potential instability)
CFS10108: Dominant period: 7.15 units (repetitive)
CFS10109: No clear repetition (potential instability)
CFS10110: Dominant period: 230.92 units (repetitive)
CFS10111: No clear repetition (potential instability)
CFS10112: Dominant period: 2.16 units (repetitive)
CFS10113: No clear repetition (potential instability)
CFS10114: Dominant period: 16.23 units (repetitive)
CFS10115: Dominant period: 4.67 units (repetitive)
CFS10116: Dominant period: 37.52 units (repetitive)
CFS10117: Dominant period: 2.56 units (repetitive)
CFS10118: No clear repetition (potential instability)
CFS10119: No clear repetition (potential instability)

Groups Trends:

CSR10001: Increasing

CEM10001: Decreasing

ORH10001: Decreasing

CRM10001: Stable
ITM10001: Increasing
CIM10001: Increasing
CPM10001: Stable
QSL10001: Stable
BIM10001: Decreasing
BPR10001: Increasing
BIH10001: Decreasing
BEV10001: Increasing
CHG10001: Decreasing
CBC10001: Stable
RTN10001: Decreasing
CII10001: Decreasing
CCM10001: Increasing
CSP10001: Increasing
CFS10001: Stable

Groups Repetitions (Fourier Analysis):

CSR10001: No clear repetition (potential instability)
CEM10001: Dominant period: 2.76 units (repetitive)
ORH10001: Dominant period: 8.78 units (repetitive)
CRM10001: Dominant period: 15.80 units (repetitive)
ITM10001: No clear repetition (potential instability)
CIM10001: Dominant period: 3.19 units (repetitive)
CPM10001: Dominant period: 34.11 units (repetitive)
QSL10001: No clear repetition (potential instability)
BIM10001: Dominant period: 7.80 units (repetitive)
BPR10001: No clear repetition (potential instability)
BIH10001: No clear repetition (potential instability)
BEV10001: Dominant period: 5.72 units (repetitive)
CHG10001: No clear repetition (potential instability)
CBC10001: No clear repetition (potential instability)
RTN10001: Dominant period: 9.35 units (repetitive)
CII10001: No clear repetition (potential instability)
CCM10001: Dominant period: 2.23 units (repetitive)
CSP10001: Dominant period: 5.92 units (repetitive)
CFS10001: Dominant period: 5.89 units (repetitive)

Sub_Modules Trends:

CSR10000: Increasing
CEM10000: Decreasing

ORH10000: Decreasing
CRM10000: Stable
ITM10000: Increasing
CIM10000: Increasing
CPM10000: Stable
QSL10000: Stable
BIM10000: Decreasing
BPR10000: Increasing
BIH10000: Decreasing
BEV10000: Increasing
CHG10000: Decreasing
CBC10000: Stable
RTN10000: Decreasing
CII10000: Decreasing
CCM10000: Increasing
CSP10000: Increasing
CFS10000: Stable

Sub_Modules Repetitions (Fourier Analysis):

CSR10000: No clear repetition (potential instability)
CEM10000: Dominant period: 2.76 units (repetitive)
ORH10000: Dominant period: 8.78 units (repetitive)
CRM10000: Dominant period: 15.80 units (repetitive)
ITM10000: No clear repetition (potential instability)
CIM10000: Dominant period: 3.19 units (repetitive)
CPM10000: Dominant period: 34.11 units (repetitive)
QSL10000: No clear repetition (potential instability)
BIM10000: Dominant period: 7.80 units (repetitive)
BPR10000: No clear repetition (potential instability)
BIH10000: No clear repetition (potential instability)
BEV10000: Dominant period: 5.72 units (repetitive)
CHG10000: No clear repetition (potential instability)
CBC10000: No clear repetition (potential instability)
RTN10000: Dominant period: 9.35 units (repetitive)
CII10000: No clear repetition (potential instability)
CCM10000: Dominant period: 2.23 units (repetitive)
CSP10000: Dominant period: 5.92 units (repetitive)
CFS10000: Dominant period: 5.89 units (repetitive)

metric_id	mean	sigma	UCL	LCL	signals	RPN	analysis_date
BEV10101	59.06	23.22	128.72	-10.59	0	35	2026-01-07
BEV10102	59.29	23.84	130.8	-12.22	0	35	2026-01-07
BEV10103	59.39	23.2	128.99	-10.2	0	35	2026-01-07
BEV10104	59.2	22.97	128.11	-9.72	0	35	2026-01-07
BEV10105	59.39	23.75	130.64	-11.87	0	35	2026-01-07
BEV10106	59.22	23.44	129.53	-11.1	0	35	2026-01-07
BEV10107	59.33	23.48	129.77	-11.1	0	35	2026-01-07
BEV10108	59.84	23.29	129.7	-10.02	0	35	2026-01-07
BEV10109	59.34	23.48	129.8	-11.11	0	35	2026-01-07
BEV10110	59.49	23.61	130.33	-11.34	0	35	2026-01-07
BEV10111	59.81	23.47	130.23	-10.62	0	35	2026-01-07
BEV10112	59.52	23.87	131.12	-12.08	0	35	2026-01-07
BEV10113	60.35	22.7	128.43	-7.74	0	35	2026-01-07
BEV10114	59.99	23.42	130.26	-10.28	0	35	2026-01-07
BEV10115	59.88	23.45	130.24	-10.48	0	35	2026-01-07
BEV10116	58.99	23.78	130.33	-12.35	0	35	2026-01-07
BEV10117	59.41	23.79	130.78	-11.96	0	35	2026-01-07
BEV10118	60.05	23.88	131.67	-11.58	0	35	2026-01-07
BEV10119	58.94	23.8	130.33	-12.45	0	35	2026-01-07
BIH10101	58.97	23.73	130.17	-12.23	0	35	2026-01-07
BIH10102	59.28	23.45	129.64	-11.08	0	35	2026-01-07
BIH10103	59.11	23.47	129.52	-11.3	0	35	2026-01-07
BIH10104	59.15	23.98	131.09	-12.79	0	35	2026-01-07
BIH10105	60.17	23.59	130.94	-10.6	0	35	2026-01-07
BIH10106	59.53	23.99	131.51	-12.45	0	35	2026-01-07
BIH10107	59.42	23.68	130.45	-11.61	0	35	2026-01-07
BIH10108	59.55	24.13	131.95	-12.85	0	35	2026-01-07
BIH10109	60.09	23.71	131.22	-11.05	0	35	2026-01-07
BIH10110	60.24	23.65	131.2	-10.72	0	35	2026-01-07
BIH10111	59.12	23.29	129.0	-10.76	0	35	2026-01-07
BIH10112	59.41	23.99	131.38	-12.55	0	35	2026-01-07
BIH10113	59.19	23.73	130.37	-12.0	0	35	2026-01-07
BIH10114	59.66	23.87	131.26	-11.93	0	35	2026-01-07
BIH10115	59.42	23.62	130.27	-11.42	0	35	2026-01-07
BIH10116	59.32	24.08	131.56	-12.92	0	35	2026-01-07
BIH10117	59.31	24.19	131.89	-13.27	0	35	2026-01-07
BIH10118	60.08	24.39	133.25	-13.1	0	35	2026-01-07
BIH10119	58.78	23.43	129.06	-11.51	0	35	2026-01-07
BIM10101	58.84	23.58	129.57	-11.9	0	35	2026-01-07
BIM10102	60.01	23.42	130.27	-10.24	0	35	2026-01-07
BIM10103	59.43	24.15	131.88	-13.01	0	35	2026-01-07
BIM10104	59.75	23.38	129.89	-10.38	0	35	2026-01-07
BIM10105	59.1	23.98	131.02	-12.83	0	35	2026-01-07
BIM10106	59.39	24.23	132.08	-13.29	0	35	2026-01-07
BIM10107	60.09	23.32	130.04	-9.87	0	35	2026-01-07
BIM10108	59.37	23.97	131.27	-12.53	0	35	2026-01-07
BIM10109	59.54	23.25	129.31	-10.22	0	35	2026-01-07
BIM10110	58.69	22.7	126.78	-9.39	0	35	2026-01-07
BIM10111	59.27	23.61	130.1	-11.57	0	35	2026-01-07
BIM10112	59.09	23.53	129.67	-11.48	0	35	2026-01-07
BIM10113	59.82	23.7	130.91	-11.28	0	35	2026-01-07
BIM10114	59.66	23.83	131.14	-11.83	0	35	2026-01-07
BIM10115	59.47	23.48	129.91	-10.98	0	35	2026-01-07
BIM10116	59.77	23.45	130.13	-10.59	0	35	2026-01-07
BIM10117	59.33	23.46	129.7	-11.04	0	35	2026-01-07
BIM10118	59.32	24.44	132.63	-13.99	0	35	2026-01-07
BIM10119	59.06	24.02	131.13	-13.02	0	35	2026-01-07
BPR10101	58.91	23.45	129.27	-11.45	0	35	2026-01-07
BPR10102	59.44	23.11	128.78	-9.9	0	35	2026-01-07
BPR10103	59.37	23.99	131.33	-12.59	0	35	2026-01-07
BPR10104	59.93	23.57	130.63	-10.77	0	35	2026-01-07
BPR10105	59.36	23.49	129.84	-11.11	0	35	2026-01-07
BPR10106	59.71	23.66	130.7	-11.27	0	35	2026-01-07
BPR10107	59.7	23.02	128.77	-9.37	0	35	2026-01-07
BPR10108	59.87	24.18	132.4	-12.66	0	35	2026-01-07
BPR10109	59.2	23.48	129.63	-11.23	0	35	2026-01-07
BPR10110	59.58	23.62	130.45	-11.29	0	35	2026-01-07
BPR10111	59.34	23.62	130.21	-11.53	0	35	2026-01-07
BPR10112	59.16	23.28	129.0	-10.68	0	35	2026-01-07
BPR10113	59.38	23.57	130.1	-11.34	0	35	2026-01-07
BPR10114	59.6	23.21	129.23	-10.03	0	35	2026-01-07
BPR10115	59.82	23.97	131.74	-12.1	0	35	2026-01-07
BPR10116	59.14	24.25	131.88	-13.6	0	35	2026-01-07
BPR10117	59.31	23.7	130.41	-11.79	0	35	2026-01-07

BPR10118	59.97	23.34	129.99	-10.05	0	35	2026-01-07
BPR10119	59.57	24.0	131.56	-12.42	0	35	2026-01-07
CBC10101	59.78	24.13	132.18	-12.62	0	35	2026-01-07
CBC10102	59.45	23.29	129.31	-10.41	0	35	2026-01-07
CBC10103	59.84	23.44	130.14	-10.47	0	35	2026-01-07
CBC10104	59.34	23.16	128.82	-10.14	0	35	2026-01-07
CBC10105	59.17	23.64	130.09	-11.75	0	35	2026-01-07
CBC10106	59.42	23.28	129.26	-10.43	0	35	2026-01-07
CBC10107	59.89	23.94	131.7	-11.92	0	35	2026-01-07
CBC10108	59.27	23.66	130.25	-11.71	0	35	2026-01-07
CBC10109	59.31	23.99	131.28	-12.66	0	35	2026-01-07
CBC10110	60.45	23.52	131.02	-10.11	0	35	2026-01-07
CBC10111	60.0	23.62	130.87	-10.88	0	35	2026-01-07
CBC10112	59.16	23.66	130.14	-11.83	0	35	2026-01-07
CBC10113	60.15	23.63	131.06	-10.75	0	35	2026-01-07
CBC10114	59.59	24.06	131.76	-12.58	0	35	2026-01-07
CBC10115	59.38	23.37	129.48	-10.72	0	35	2026-01-07
CBC10116	59.01	23.63	129.88	-11.87	0	35	2026-01-07
CBC10117	59.14	23.27	128.94	-10.66	0	35	2026-01-07
CBC10118	59.16	23.56	129.83	-11.5	0	35	2026-01-07
CBC10119	59.54	23.75	130.78	-11.7	0	35	2026-01-07
CCM10101	59.97	23.21	129.58	-9.65	0	35	2026-01-07
CCM10102	59.39	23.87	131.02	-12.23	0	35	2026-01-07
CCM10103	59.67	23.58	130.4	-11.07	0	35	2026-01-07
CCM10104	59.28	23.77	130.58	-12.03	0	35	2026-01-07
CCM10105	59.24	23.82	130.69	-12.21	0	35	2026-01-07
CCM10106	59.06	23.09	128.32	-10.19	0	35	2026-01-07
CCM10107	58.71	24.23	131.4	-13.97	0	35	2026-01-07
CCM10108	59.8	23.93	131.6	-11.99	0	35	2026-01-07
CCM10109	59.1	23.8	130.51	-12.31	0	35	2026-01-07
CCM10110	59.84	23.26	129.61	-9.93	0	35	2026-01-07
CCM10111	60.16	23.7	131.26	-10.94	0	35	2026-01-07
CCM10112	59.4	23.6	130.2	-11.41	0	35	2026-01-07
CCM10113	59.33	23.67	130.33	-11.67	0	35	2026-01-07
CCM10114	59.53	23.93	131.31	-12.25	0	35	2026-01-07
CCM10115	60.29	23.64	131.21	-10.63	0	35	2026-01-07
CCM10116	60.42	23.22	130.07	-9.24	0	35	2026-01-07
CCM10117	59.73	24.02	131.79	-12.32	0	35	2026-01-07
CCM10118	59.33	23.67	130.34	-11.68	0	35	2026-01-07
CCM10119	60.19	23.62	131.06	-10.68	0	35	2026-01-07
CEM10101	59.91	23.24	129.63	-9.8	0	35	2026-01-07
CEM10102	59.39	23.83	130.87	-12.09	0	35	2026-01-07
CEM10103	58.81	23.56	129.5	-11.87	0	35	2026-01-07
CEM10104	59.39	23.26	129.18	-10.4	0	35	2026-01-07
CEM10105	59.46	23.81	130.87	-11.96	0	35	2026-01-07
CEM10106	59.96	23.54	130.59	-10.67	0	35	2026-01-07
CEM10107	58.52	22.84	127.04	-10.01	0	35	2026-01-07
CEM10108	59.32	23.43	129.6	-10.96	0	35	2026-01-07
CEM10109	59.19	23.65	130.15	-11.77	0	35	2026-01-07
CEM10110	59.43	23.25	129.17	-10.31	0	35	2026-01-07
CEM10111	59.8	23.39	129.96	-10.36	0	35	2026-01-07
CEM10112	59.44	23.98	131.38	-12.5	0	35	2026-01-07
CEM10113	59.01	23.25	128.76	-10.74	0	35	2026-01-07
CEM10114	59.03	23.63	129.91	-11.84	0	35	2026-01-07
CFS10101	59.34	23.47	129.75	-11.07	0	35	2026-01-07
CFS10102	59.36	23.5	129.85	-11.14	0	35	2026-01-07
CFS10103	59.91	23.34	129.94	-10.12	0	35	2026-01-07
CFS10104	59.4	23.98	131.34	-12.54	0	35	2026-01-07
CFS10105	59.88	23.72	131.03	-11.28	0	35	2026-01-07
CFS10106	59.28	23.67	130.29	-11.73	0	35	2026-01-07
CFS10107	59.61	23.07	128.82	-9.6	0	35	2026-01-07
CFS10108	58.76	23.54	129.39	-11.86	0	35	2026-01-07
CFS10109	59.81	23.15	129.26	-9.64	0	35	2026-01-07
CFS10110	59.03	23.21	128.66	-10.59	0	35	2026-01-07
CFS10111	59.06	24.11	131.39	-13.27	0	35	2026-01-07
CFS10112	59.62	23.62	130.47	-11.22	0	35	2026-01-07
CFS10113	59.64	24.3	132.52	-13.25	0	35	2026-01-07
CFS10114	59.65	23.67	130.65	-11.35	0	35	2026-01-07
CFS10115	59.6	22.81	128.04	-8.84	0	35	2026-01-07
CFS10116	59.45	23.58	130.18	-11.28	0	35	2026-01-07
CFS10117	59.32	23.77	130.64	-12.0	0	35	2026-01-07
CFS10118	59.81	22.84	128.33	-8.72	0	35	2026-01-07
CFS10119	59.63	23.56	130.31	-11.05	0	35	2026-01-07
CHG10101	59.75	23.87	131.36	-11.86	0	35	2026-01-07
CHG10102	59.3	23.33	129.28	-10.68	0	35	2026-01-07

CHG10103	59.99	23.81	131.43	-11.45	0	35	2026-01-07
CHG10104	59.69	23.47	130.11	-10.74	0	35	2026-01-07
CHG10105	59.22	23.9	130.91	-12.47	0	35	2026-01-07
CHG10106	58.53	23.65	129.48	-12.41	0	35	2026-01-07
CHG10107	59.27	23.9	130.98	-12.43	0	35	2026-01-07
CHG10108	59.95	24.37	133.05	-13.15	0	35	2026-01-07
CHG10109	59.09	23.93	130.87	-12.7	0	35	2026-01-07
CHG10110	59.5	23.23	129.19	-10.19	0	35	2026-01-07
CHG10111	59.04	23.54	129.67	-11.6	0	35	2026-01-07
CHG10112	59.17	23.31	129.1	-10.76	0	35	2026-01-07
CHG10113	58.97	23.57	129.68	-11.74	0	35	2026-01-07
CHG10114	59.17	23.84	130.69	-12.34	0	35	2026-01-07
CHG10115	59.11	23.46	129.48	-11.26	0	35	2026-01-07
CHG10116	60.09	23.72	131.24	-11.06	0	35	2026-01-07
CHG10117	59.12	23.91	130.86	-12.61	0	35	2026-01-07
CHG10118	59.48	23.51	130.01	-11.05	0	35	2026-01-07
CHG10119	59.58	23.79	130.95	-11.79	0	35	2026-01-07
CII10101	60.39	23.93	132.19	-11.41	0	35	2026-01-07
CII10102	59.76	23.22	129.43	-9.9	0	35	2026-01-07
CII10103	58.84	23.66	129.82	-12.13	0	35	2026-01-07
CII10104	59.62	24.05	131.77	-12.53	0	35	2026-01-07
CII10105	59.43	24.11	131.77	-12.92	0	35	2026-01-07
CII10106	59.26	23.55	129.93	-11.4	0	35	2026-01-07
CII10107	59.69	23.37	129.8	-10.41	0	35	2026-01-07
CII10108	59.41	23.27	129.22	-10.4	0	35	2026-01-07
CII10109	59.84	23.72	131.0	-11.32	0	35	2026-01-07
CII10110	58.94	23.39	129.11	-11.23	0	35	2026-01-07
CII10111	59.72	23.86	131.3	-11.86	0	35	2026-01-07
CII10112	60.18	23.52	130.73	-10.38	0	35	2026-01-07
CII10113	59.71	23.21	129.35	-9.93	0	35	2026-01-07
CII10114	60.02	23.89	131.69	-11.66	0	35	2026-01-07
CII10115	59.63	23.91	131.35	-12.09	0	35	2026-01-07
CII10116	59.29	23.5	129.78	-11.21	0	35	2026-01-07
CII10117	59.2	23.68	130.25	-11.85	0	35	2026-01-07
CII10118	60.0	23.39	130.17	-10.17	0	35	2026-01-07
CII10119	59.36	23.8	130.77	-12.04	0	35	2026-01-07
CIM10101	59.65	23.73	130.84	-11.54	0	35	2026-01-07
CIM10102	59.19	23.4	129.4	-11.02	0	35	2026-01-07
CIM10103	59.42	23.07	128.62	-9.78	0	35	2026-01-07
CIM10104	59.87	23.82	131.32	-11.59	0	35	2026-01-07
CIM10105	59.24	23.82	130.71	-12.23	0	35	2026-01-07
CIM10106	59.75	24.05	131.91	-12.42	0	35	2026-01-07
CIM10107	59.0	23.34	129.03	-11.02	0	35	2026-01-07
CIM10108	59.85	24.36	132.92	-13.22	0	35	2026-01-07
CIM10109	59.9	23.92	131.65	-11.85	0	35	2026-01-07
CIM10110	59.88	23.71	131.01	-11.26	0	35	2026-01-07
CIM10111	59.78	24.21	132.4	-12.84	0	35	2026-01-07
CIM10112	59.48	23.67	130.49	-11.53	0	35	2026-01-07
CIM10113	60.69	23.44	131.03	-9.64	0	35	2026-01-07
CIM10114	59.38	23.12	128.74	-9.98	0	35	2026-01-07
CIM10115	59.79	23.59	130.57	-10.99	0	35	2026-01-07
CIM10116	59.43	24.26	132.21	-13.35	0	35	2026-01-07
CIM10117	60.04	23.33	130.04	-9.95	0	35	2026-01-07
CIM10118	59.53	24.01	131.55	-12.49	0	35	2026-01-07
CIM10119	58.73	23.72	129.9	-12.43	0	35	2026-01-07
CPM10101	58.88	23.43	129.16	-11.4	0	35	2026-01-07
CPM10102	59.12	24.02	131.2	-12.95	0	35	2026-01-07
CPM10103	59.96	23.76	131.23	-11.31	0	35	2026-01-07
CPM10104	59.82	24.1	132.11	-12.46	0	35	2026-01-07
CPM10105	59.1	23.8	130.51	-12.31	0	35	2026-01-07
CPM10106	59.46	23.88	131.08	-12.17	0	35	2026-01-07
CPM10107	59.81	23.86	131.4	-11.78	0	35	2026-01-07
CPM10108	59.62	22.93	128.42	-9.18	0	35	2026-01-07
CPM10109	59.41	23.32	129.37	-10.56	0	35	2026-01-07
CPM10110	59.88	23.24	129.61	-9.85	0	35	2026-01-07
CPM10111	59.27	23.75	130.51	-11.97	0	35	2026-01-07
CPM10112	59.54	23.85	131.09	-12.02	0	35	2026-01-07
CPM10113	59.79	23.71	130.92	-11.34	0	35	2026-01-07
CPM10114	59.63	23.51	130.15	-10.89	0	35	2026-01-07
CPM10115	59.38	23.49	129.85	-11.09	0	35	2026-01-07
CPM10116	59.43	23.9	131.13	-12.26	0	35	2026-01-07
CPM10117	58.63	23.92	130.38	-13.11	0	35	2026-01-07
CPM10118	59.49	23.17	129.0	-10.02	0	35	2026-01-07
CPM10119	59.58	23.55	130.23	-11.06	0	35	2026-01-07
CRM10101	59.76	24.21	132.39	-12.87	0	35	2026-01-07

CRM10102	59.11	23.77	130.42	-12.2	0	35	2026-01-07
CRM10103	59.33	23.59	130.11	-11.44	0	35	2026-01-07
CRM10104	59.49	23.81	130.93	-11.94	0	35	2026-01-07
CRM10105	59.51	23.54	130.13	-11.1	0	35	2026-01-07
CRM10106	59.2	23.92	130.97	-12.57	0	35	2026-01-07
CRM10107	59.39	23.35	129.45	-10.68	0	35	2026-01-07
CRM10108	59.11	23.3	129.02	-10.8	0	35	2026-01-07
CRM10109	58.85	23.6	129.65	-11.95	0	35	2026-01-07
CRM10110	59.59	23.35	129.65	-10.48	0	35	2026-01-07
CRM10111	60.05	23.6	130.84	-10.74	0	35	2026-01-07
CRM10112	58.73	24.27	131.53	-14.06	0	35	2026-01-07
CRM10113	59.45	23.28	129.31	-10.4	0	35	2026-01-07
CRM10114	59.63	23.77	130.93	-11.67	0	35	2026-01-07
CRM10115	59.07	23.62	129.91	-11.78	0	35	2026-01-07
CRM10116	59.5	23.49	129.97	-10.96	0	35	2026-01-07
CRM10117	59.49	22.71	127.62	-8.64	0	35	2026-01-07
CRM10118	59.56	23.86	131.16	-12.03	0	35	2026-01-07
CRM10119	60.01	24.44	133.32	-13.3	0	35	2026-01-07
CSP10101	60.3	23.57	131.02	-10.41	0	35	2026-01-07
CSP10102	59.69	23.69	130.76	-11.37	0	35	2026-01-07
CSP10103	59.98	23.14	129.4	-9.44	0	35	2026-01-07
CSP10104	59.5	24.18	132.02	-13.03	0	35	2026-01-07
CSP10105	58.37	23.78	129.73	-12.98	0	35	2026-01-07
CSP10106	58.92	23.68	129.97	-12.13	0	35	2026-01-07
CSP10107	60.01	23.7	131.1	-11.08	0	35	2026-01-07
CSP10108	60.04	23.97	131.95	-11.88	0	35	2026-01-07
CSP10109	59.73	24.18	132.28	-12.82	0	35	2026-01-07
CSP10110	59.17	23.38	129.3	-10.95	0	35	2026-01-07
CSP10111	60.2	23.57	130.92	-10.52	0	35	2026-01-07
CSP10112	58.88	23.7	129.98	-12.22	0	35	2026-01-07
CSP10113	59.71	23.62	130.58	-11.16	0	35	2026-01-07
CSP10114	58.54	23.22	128.21	-11.12	0	35	2026-01-07
CSP10115	60.24	23.31	130.15	-9.68	0	35	2026-01-07
CSP10116	58.98	23.83	130.46	-12.5	0	35	2026-01-07
CSP10117	59.14	23.65	130.08	-11.81	0	35	2026-01-07
CSP10118	60.24	23.75	131.5	-11.02	0	35	2026-01-07
CSP10119	59.78	24.05	131.92	-12.37	0	35	2026-01-07
CSR10101	59.55	23.72	130.69	-11.6	0	35	2026-01-07
CSR10102	59.08	23.54	129.7	-11.53	0	35	2026-01-07
CSR10103	59.4	23.84	130.9	-12.11	0	35	2026-01-07
CSR10104	59.54	23.26	129.31	-10.22	0	35	2026-01-07
CSR10105	59.9	23.51	130.44	-10.64	0	35	2026-01-07
CSR10106	58.98	23.84	130.51	-12.54	0	35	2026-01-07
CSR10107	59.35	23.94	131.18	-12.47	0	35	2026-01-07
CSR10108	59.42	23.89	131.08	-12.24	0	35	2026-01-07
CSR10109	60.11	23.22	129.77	-9.55	0	35	2026-01-07
CSR10110	58.91	23.48	129.35	-11.52	0	35	2026-01-07
CSR10111	59.8	23.69	130.87	-11.27	0	35	2026-01-07
CSR10112	59.53	23.48	129.97	-10.91	0	35	2026-01-07
CSR10113	59.44	23.45	129.79	-10.9	0	35	2026-01-07
CSR10114	59.07	23.21	128.7	-10.57	0	35	2026-01-07
CSR10115	59.72	23.69	130.8	-11.35	0	35	2026-01-07
ITM10101	59.01	23.65	129.96	-11.94	0	35	2026-01-07
ITM10102	60.03	24.03	132.11	-12.05	0	35	2026-01-07
ITM10103	59.46	23.68	130.5	-11.58	0	35	2026-01-07
ITM10104	59.39	23.8	130.8	-12.02	0	35	2026-01-07
ITM10105	58.65	23.86	130.23	-12.94	0	35	2026-01-07
ITM10106	59.31	23.62	130.19	-11.56	0	35	2026-01-07
ITM10107	59.87	23.36	129.95	-10.22	0	35	2026-01-07
ITM10108	59.36	24.13	131.75	-13.02	0	35	2026-01-07
ITM10109	59.38	23.88	131.03	-12.26	0	35	2026-01-07
ITM10110	59.29	24.01	131.31	-12.73	0	35	2026-01-07
ITM10111	59.01	23.89	130.67	-12.64	0	35	2026-01-07
ITM10112	59.37	23.64	130.29	-11.56	0	35	2026-01-07
ITM10113	59.04	23.16	128.53	-10.45	0	35	2026-01-07
ITM10114	58.47	23.84	129.99	-13.05	0	35	2026-01-07
ITM10115	58.52	23.91	130.25	-13.21	0	35	2026-01-07
ITM10116	59.4	24.14	131.83	-13.03	0	35	2026-01-07
ITM10117	59.83	24.11	132.16	-12.5	0	35	2026-01-07
ITM10118	58.74	23.62	129.59	-12.11	0	35	2026-01-07
ITM10119	60.27	23.85	131.83	-11.3	0	35	2026-01-07
ORH10101	59.05	23.61	129.9	-11.79	0	35	2026-01-07
ORH10102	59.9	23.63	130.78	-10.99	0	35	2026-01-07
ORH10103	58.17	23.58	128.91	-12.57	0	35	2026-01-07
ORH10104	59.15	23.4	129.36	-11.07	0	35	2026-01-07

ORH10105	59.77	23.61	130.6	-11.07	0	35	2026-01-07
ORH10106	59.75	23.41	129.98	-10.47	0	35	2026-01-07
ORH10107	60.58	23.56	131.27	-10.11	0	35	2026-01-07
ORH10108	60.14	23.34	130.17	-9.88	0	35	2026-01-07
ORH10109	59.57	23.2	129.17	-10.03	0	35	2026-01-07
ORH10110	58.98	24.03	131.08	-13.12	0	35	2026-01-07
ORH10111	59.76	23.37	129.86	-10.35	0	35	2026-01-07
ORH10112	59.68	23.19	129.26	-9.9	0	35	2026-01-07
ORH10113	59.68	23.77	130.97	-11.62	0	35	2026-01-07
ORH10114	59.56	23.02	128.64	-9.51	0	35	2026-01-07
ORH10115	59.7	23.58	130.45	-11.05	0	35	2026-01-07
QSL10101	59.73	23.39	129.9	-10.44	0	35	2026-01-07
QSL10102	59.15	23.86	130.72	-12.42	0	35	2026-01-07
QSL10103	59.37	23.95	131.21	-12.47	0	35	2026-01-07
QSL10104	59.0	23.58	129.74	-11.74	0	35	2026-01-07
QSL10105	59.47	23.21	129.11	-10.16	0	35	2026-01-07
QSL10106	59.2	23.69	130.28	-11.88	0	35	2026-01-07
QSL10107	59.48	23.67	130.49	-11.52	0	35	2026-01-07
QSL10108	60.04	23.53	130.62	-10.54	0	35	2026-01-07
QSL10109	59.18	23.46	129.55	-11.2	0	35	2026-01-07
QSL10110	59.74	24.04	131.85	-12.37	0	35	2026-01-07
QSL10111	59.31	23.75	130.57	-11.96	0	35	2026-01-07
QSL10112	60.42	23.91	132.15	-11.32	0	35	2026-01-07
QSL10113	60.79	23.37	130.89	-9.31	0	35	2026-01-07
QSL10114	59.23	23.71	130.35	-11.88	0	35	2026-01-07
QSL10115	59.53	23.42	129.78	-10.72	0	35	2026-01-07
QSL10116	59.43	23.1	128.72	-9.86	0	35	2026-01-07
QSL10117	59.34	23.1	128.64	-9.95	0	35	2026-01-07
QSL10118	59.25	23.58	130.0	-11.5	0	35	2026-01-07
QSL10119	58.5	23.4	128.69	-11.69	0	35	2026-01-07
RTN10101	58.95	24.11	131.29	-13.38	0	35	2026-01-07
RTN10102	59.52	23.25	129.26	-10.22	0	35	2026-01-07
RTN10103	58.99	23.65	129.96	-11.97	0	35	2026-01-07
RTN10104	59.61	23.7	130.71	-11.49	0	35	2026-01-07
RTN10105	59.25	23.31	129.17	-10.67	0	35	2026-01-07
RTN10106	59.55	23.74	130.77	-11.67	0	35	2026-01-07
RTN10107	59.85	23.48	130.3	-10.59	0	35	2026-01-07
RTN10108	59.71	22.98	128.64	-9.23	0	35	2026-01-07
RTN10109	59.31	24.25	132.07	-13.44	0	35	2026-01-07
RTN10110	59.61	23.6	130.4	-11.18	0	35	2026-01-07
RTN10111	59.97	23.38	130.11	-10.17	0	35	2026-01-07
RTN10112	59.08	23.45	129.42	-11.26	0	35	2026-01-07
RTN10113	60.1	24.31	133.03	-12.83	0	35	2026-01-07
RTN10114	59.23	24.36	132.32	-13.87	0	35	2026-01-07
RTN10115	59.49	23.71	130.62	-11.64	0	35	2026-01-07
RTN10116	59.6	24.4	132.8	-13.59	0	35	2026-01-07
RTN10117	59.27	23.27	129.08	-10.55	0	35	2026-01-07
RTN10118	59.47	23.34	129.5	-10.55	0	35	2026-01-07
RTN10119	59.41	23.42	129.68	-10.86	0	35	2026-01-07

=== rp_n_summary_customer.txt ===

metric_id	mean	sigma	UCL	LCL	signals	RPN	analysis_date
BEV10101	59.06	23.22	128.72	-10.59	0	35	2026-01-07
BEV10102	59.29	23.84	130.8	-12.22	0	35	2026-01-07
BEV10103	59.39	23.2	128.99	-10.2	0	35	2026-01-07
BEV10104	59.2	22.97	128.11	-9.72	0	35	2026-01-07
BEV10105	59.39	23.75	130.64	-11.87	0	35	2026-01-07
BEV10106	59.22	23.44	129.53	-11.1	0	35	2026-01-07
BEV10107	59.33	23.48	129.77	-11.1	0	35	2026-01-07
BEV10108	59.84	23.29	129.7	-10.02	0	35	2026-01-07
BEV10109	59.34	23.48	129.8	-11.11	0	35	2026-01-07
BEV10110	59.49	23.61	130.33	-11.34	0	35	2026-01-07
BEV10111	59.81	23.47	130.23	-10.62	0	35	2026-01-07
BEV10112	59.52	23.87	131.12	-12.08	0	35	2026-01-07
BEV10113	60.35	22.7	128.43	-7.74	0	35	2026-01-07
BEV10114	59.99	23.42	130.26	-10.28	0	35	2026-01-07
BEV10115	59.88	23.45	130.24	-10.48	0	35	2026-01-07
BEV10116	58.99	23.78	130.33	-12.35	0	35	2026-01-07
BEV10117	59.41	23.79	130.78	-11.96	0	35	2026-01-07
BEV10118	60.05	23.88	131.67	-11.58	0	35	2026-01-07
BEV10119	58.94	23.8	130.33	-12.45	0	35	2026-01-07
BIH10101	58.97	23.73	130.17	-12.23	0	35	2026-01-07

BIH10102	59.28	23.45	129.64	-11.08	0	35	2026-01-07
BIH10103	59.11	23.47	129.52	-11.3	0	35	2026-01-07
BIH10104	59.15	23.98	131.09	-12.79	0	35	2026-01-07
BIH10105	60.17	23.59	130.94	-10.6	0	35	2026-01-07
BIH10106	59.53	23.99	131.51	-12.45	0	35	2026-01-07
BIH10107	59.42	23.68	130.45	-11.61	0	35	2026-01-07
BIH10108	59.55	24.13	131.95	-12.85	0	35	2026-01-07
BIH10109	60.09	23.71	131.22	-11.05	0	35	2026-01-07
BIH10110	60.24	23.65	131.2	-10.72	0	35	2026-01-07
BIH10111	59.12	23.29	129.0	-10.76	0	35	2026-01-07
BIH10112	59.41	23.99	131.38	-12.55	0	35	2026-01-07
BIH10113	59.19	23.73	130.37	-12.0	0	35	2026-01-07
BIH10114	59.66	23.87	131.26	-11.93	0	35	2026-01-07
BIH10115	59.42	23.62	130.27	-11.42	0	35	2026-01-07
BIH10116	59.32	24.08	131.56	-12.92	0	35	2026-01-07
BIH10117	59.31	24.19	131.89	-13.27	0	35	2026-01-07
BIH10118	60.08	24.39	133.25	-13.1	0	35	2026-01-07
BIH10119	58.78	23.43	129.06	-11.51	0	35	2026-01-07
BIM10101	58.84	23.58	129.57	-11.9	0	35	2026-01-07
BIM10102	60.01	23.42	130.27	-10.24	0	35	2026-01-07
BIM10103	59.43	24.15	131.88	-13.01	0	35	2026-01-07
BIM10104	59.75	23.38	129.89	-10.38	0	35	2026-01-07
BIM10105	59.1	23.98	131.02	-12.83	0	35	2026-01-07
BIM10106	59.39	24.23	132.08	-13.29	0	35	2026-01-07
BIM10107	60.09	23.32	130.04	-9.87	0	35	2026-01-07
BIM10108	59.37	23.97	131.27	-12.53	0	35	2026-01-07
BIM10109	59.54	23.25	129.31	-10.22	0	35	2026-01-07
BIM10110	58.69	22.7	126.78	-9.39	0	35	2026-01-07
BIM10111	59.27	23.61	130.1	-11.57	0	35	2026-01-07
BIM10112	59.09	23.53	129.67	-11.48	0	35	2026-01-07
BIM10113	59.82	23.7	130.91	-11.28	0	35	2026-01-07
BIM10114	59.66	23.83	131.14	-11.83	0	35	2026-01-07
BIM10115	59.47	23.48	129.91	-10.98	0	35	2026-01-07
BIM10116	59.77	23.45	130.13	-10.59	0	35	2026-01-07
BIM10117	59.33	23.46	129.7	-11.04	0	35	2026-01-07
BIM10118	59.32	24.44	132.63	-13.99	0	35	2026-01-07
BIM10119	59.06	24.02	131.13	-13.02	0	35	2026-01-07
BPR10101	58.91	23.45	129.27	-11.45	0	35	2026-01-07
BPR10102	59.44	23.11	128.78	-9.9	0	35	2026-01-07
BPR10103	59.37	23.99	131.33	-12.59	0	35	2026-01-07
BPR10104	59.93	23.57	130.63	-10.77	0	35	2026-01-07
BPR10105	59.36	23.49	129.84	-11.11	0	35	2026-01-07
BPR10106	59.71	23.66	130.7	-11.27	0	35	2026-01-07
BPR10107	59.7	23.02	128.77	-9.37	0	35	2026-01-07
BPR10108	59.87	24.18	132.4	-12.66	0	35	2026-01-07
BPR10109	59.2	23.48	129.63	-11.23	0	35	2026-01-07
BPR10110	59.58	23.62	130.45	-11.29	0	35	2026-01-07
BPR10111	59.34	23.62	130.21	-11.53	0	35	2026-01-07
BPR10112	59.16	23.28	129.0	-10.68	0	35	2026-01-07
BPR10113	59.38	23.57	130.1	-11.34	0	35	2026-01-07
BPR10114	59.6	23.21	129.23	-10.03	0	35	2026-01-07
BPR10115	59.82	23.97	131.74	-12.1	0	35	2026-01-07
BPR10116	59.14	24.25	131.88	-13.6	0	35	2026-01-07
BPR10117	59.31	23.7	130.41	-11.79	0	35	2026-01-07
BPR10118	59.97	23.34	129.99	-10.05	0	35	2026-01-07
BPR10119	59.57	24.0	131.56	-12.42	0	35	2026-01-07
CBC10101	59.78	24.13	132.18	-12.62	0	35	2026-01-07
CBC10102	59.45	23.29	129.31	-10.41	0	35	2026-01-07
CBC10103	59.84	23.44	130.14	-10.47	0	35	2026-01-07
CBC10104	59.34	23.16	128.82	-10.14	0	35	2026-01-07
CBC10105	59.17	23.64	130.09	-11.75	0	35	2026-01-07
CBC10106	59.42	23.28	129.26	-10.43	0	35	2026-01-07
CBC10107	59.89	23.94	131.7	-11.92	0	35	2026-01-07
CBC10108	59.27	23.66	130.25	-11.71	0	35	2026-01-07
CBC10109	59.31	23.99	131.28	-12.66	0	35	2026-01-07
CBC10110	60.45	23.52	131.02	-10.11	0	35	2026-01-07
CBC10111	60.0	23.62	130.87	-10.88	0	35	2026-01-07
CBC10112	59.16	23.66	130.14	-11.83	0	35	2026-01-07
CBC10113	60.15	23.63	131.06	-10.75	0	35	2026-01-07
CBC10114	59.59	24.06	131.76	-12.58	0	35	2026-01-07
CBC10115	59.38	23.37	129.48	-10.72	0	35	2026-01-07
CBC10116	59.01	23.63	129.88	-11.87	0	35	2026-01-07
CBC10117	59.14	23.27	128.94	-10.66	0	35	2026-01-07
CBC10118	59.16	23.56	129.83	-11.5	0	35	2026-01-07
CBC10119	59.54	23.75	130.78	-11.7	0	35	2026-01-07

CCM10101	59.97	23.21	129.58	-9.65	0	35	2026-01-07
CCM10102	59.39	23.87	131.02	-12.23	0	35	2026-01-07
CCM10103	59.67	23.58	130.4	-11.07	0	35	2026-01-07
CCM10104	59.28	23.77	130.58	-12.03	0	35	2026-01-07
CCM10105	59.24	23.82	130.69	-12.21	0	35	2026-01-07
CCM10106	59.06	23.09	128.32	-10.19	0	35	2026-01-07
CCM10107	58.71	24.23	131.4	-13.97	0	35	2026-01-07
CCM10108	59.8	23.93	131.6	-11.99	0	35	2026-01-07
CCM10109	59.1	23.8	130.51	-12.31	0	35	2026-01-07
CCM10110	59.84	23.26	129.61	-9.93	0	35	2026-01-07
CCM10111	60.16	23.7	131.26	-10.94	0	35	2026-01-07
CCM10112	59.4	23.6	130.2	-11.41	0	35	2026-01-07
CCM10113	59.33	23.67	130.33	-11.67	0	35	2026-01-07
CCM10114	59.53	23.93	131.31	-12.25	0	35	2026-01-07
CCM10115	60.29	23.64	131.21	-10.63	0	35	2026-01-07
CCM10116	60.42	23.22	130.07	-9.24	0	35	2026-01-07
CCM10117	59.73	24.02	131.79	-12.32	0	35	2026-01-07
CCM10118	59.33	23.67	130.34	-11.68	0	35	2026-01-07
CCM10119	60.19	23.62	131.06	-10.68	0	35	2026-01-07
CEM10101	59.91	23.24	129.63	-9.8	0	35	2026-01-07
CEM10102	59.39	23.83	130.87	-12.09	0	35	2026-01-07
CEM10103	58.81	23.56	129.5	-11.87	0	35	2026-01-07
CEM10104	59.39	23.26	129.18	-10.4	0	35	2026-01-07
CEM10105	59.46	23.81	130.87	-11.96	0	35	2026-01-07
CEM10106	59.96	23.54	130.59	-10.67	0	35	2026-01-07
CEM10107	58.52	22.84	127.04	-10.01	0	35	2026-01-07
CEM10108	59.32	23.43	129.6	-10.96	0	35	2026-01-07
CEM10109	59.19	23.65	130.15	-11.77	0	35	2026-01-07
CEM10110	59.43	23.25	129.17	-10.31	0	35	2026-01-07
CEM10111	59.8	23.39	129.96	-10.36	0	35	2026-01-07
CEM10112	59.44	23.98	131.38	-12.5	0	35	2026-01-07
CEM10113	59.01	23.25	128.76	-10.74	0	35	2026-01-07
CEM10114	59.03	23.63	129.91	-11.84	0	35	2026-01-07
CFS10101	59.34	23.47	129.75	-11.07	0	35	2026-01-07
CFS10102	59.36	23.5	129.85	-11.14	0	35	2026-01-07
CFS10103	59.91	23.34	129.94	-10.12	0	35	2026-01-07
CFS10104	59.4	23.98	131.34	-12.54	0	35	2026-01-07
CFS10105	59.88	23.72	131.03	-11.28	0	35	2026-01-07
CFS10106	59.28	23.67	130.29	-11.73	0	35	2026-01-07
CFS10107	59.61	23.07	128.82	-9.6	0	35	2026-01-07
CFS10108	58.76	23.54	129.39	-11.86	0	35	2026-01-07
CFS10109	59.81	23.15	129.26	-9.64	0	35	2026-01-07
CFS10110	59.03	23.21	128.66	-10.59	0	35	2026-01-07
CFS10111	59.06	24.11	131.39	-13.27	0	35	2026-01-07
CFS10112	59.62	23.62	130.47	-11.22	0	35	2026-01-07
CFS10113	59.64	24.3	132.52	-13.25	0	35	2026-01-07
CFS10114	59.65	23.67	130.65	-11.35	0	35	2026-01-07
CFS10115	59.6	22.81	128.04	-8.84	0	35	2026-01-07
CFS10116	59.45	23.58	130.18	-11.28	0	35	2026-01-07
CFS10117	59.32	23.77	130.64	-12.0	0	35	2026-01-07
CFS10118	59.81	22.84	128.33	-8.72	0	35	2026-01-07
CFS10119	59.63	23.56	130.31	-11.05	0	35	2026-01-07
CHG10101	59.75	23.87	131.36	-11.86	0	35	2026-01-07
CHG10102	59.3	23.33	129.28	-10.68	0	35	2026-01-07
CHG10103	59.99	23.81	131.43	-11.45	0	35	2026-01-07
CHG10104	59.69	23.47	130.11	-10.74	0	35	2026-01-07
CHG10105	59.22	23.9	130.91	-12.47	0	35	2026-01-07
CHG10106	58.53	23.65	129.48	-12.41	0	35	2026-01-07
CHG10107	59.27	23.9	130.98	-12.43	0	35	2026-01-07
CHG10108	59.95	24.37	133.05	-13.15	0	35	2026-01-07
CHG10109	59.09	23.93	130.87	-12.7	0	35	2026-01-07
CHG10110	59.5	23.23	129.19	-10.19	0	35	2026-01-07
CHG10111	59.04	23.54	129.67	-11.6	0	35	2026-01-07
CHG10112	59.17	23.31	129.1	-10.76	0	35	2026-01-07
CHG10113	58.97	23.57	129.68	-11.74	0	35	2026-01-07
CHG10114	59.17	23.84	130.69	-12.34	0	35	2026-01-07
CHG10115	59.11	23.46	129.48	-11.26	0	35	2026-01-07
CHG10116	60.09	23.72	131.24	-11.06	0	35	2026-01-07
CHG10117	59.12	23.91	130.86	-12.61	0	35	2026-01-07
CHG10118	59.48	23.51	130.01	-11.05	0	35	2026-01-07
CHG10119	59.58	23.79	130.95	-11.79	0	35	2026-01-07
CII10101	60.39	23.93	132.19	-11.41	0	35	2026-01-07
CII10102	59.76	23.22	129.43	-9.9	0	35	2026-01-07
CII10103	58.84	23.66	129.82	-12.13	0	35	2026-01-07
CII10104	59.62	24.05	131.77	-12.53	0	35	2026-01-07

CII10105	59.43	24.11	131.77	-12.92	0	35	2026-01-07
CII10106	59.26	23.55	129.93	-11.4	0	35	2026-01-07
CII10107	59.69	23.37	129.8	-10.41	0	35	2026-01-07
CII10108	59.41	23.27	129.22	-10.4	0	35	2026-01-07
CII10109	59.84	23.72	131.0	-11.32	0	35	2026-01-07
CII10110	58.94	23.39	129.11	-11.23	0	35	2026-01-07
CII10111	59.72	23.86	131.3	-11.86	0	35	2026-01-07
CII10112	60.18	23.52	130.73	-10.38	0	35	2026-01-07
CII10113	59.71	23.21	129.35	-9.93	0	35	2026-01-07
CII10114	60.02	23.89	131.69	-11.66	0	35	2026-01-07
CII10115	59.63	23.91	131.35	-12.09	0	35	2026-01-07
CII10116	59.29	23.5	129.78	-11.21	0	35	2026-01-07
CII10117	59.2	23.68	130.25	-11.85	0	35	2026-01-07
CII10118	60.0	23.39	130.17	-10.17	0	35	2026-01-07
CII10119	59.36	23.8	130.77	-12.04	0	35	2026-01-07
CIM10101	59.65	23.73	130.84	-11.54	0	35	2026-01-07
CIM10102	59.19	23.4	129.4	-11.02	0	35	2026-01-07
CIM10103	59.42	23.07	128.62	-9.78	0	35	2026-01-07
CIM10104	59.87	23.82	131.32	-11.59	0	35	2026-01-07
CIM10105	59.24	23.82	130.71	-12.23	0	35	2026-01-07
CIM10106	59.75	24.05	131.91	-12.42	0	35	2026-01-07
CIM10107	59.0	23.34	129.03	-11.02	0	35	2026-01-07
CIM10108	59.85	24.36	132.92	-13.22	0	35	2026-01-07
CIM10109	59.9	23.92	131.65	-11.85	0	35	2026-01-07
CIM10110	59.88	23.71	131.01	-11.26	0	35	2026-01-07
CIM10111	59.78	24.21	132.4	-12.84	0	35	2026-01-07
CIM10112	59.48	23.67	130.49	-11.53	0	35	2026-01-07
CIM10113	60.69	23.44	131.03	-9.64	0	35	2026-01-07
CIM10114	59.38	23.12	128.74	-9.98	0	35	2026-01-07
CIM10115	59.79	23.59	130.57	-10.99	0	35	2026-01-07
CIM10116	59.43	24.26	132.21	-13.35	0	35	2026-01-07
CIM10117	60.04	23.33	130.04	-9.95	0	35	2026-01-07
CIM10118	59.53	24.01	131.55	-12.49	0	35	2026-01-07
CIM10119	58.73	23.72	129.9	-12.43	0	35	2026-01-07
CPM10101	58.88	23.43	129.16	-11.4	0	35	2026-01-07
CPM10102	59.12	24.02	131.2	-12.95	0	35	2026-01-07
CPM10103	59.96	23.76	131.23	-11.31	0	35	2026-01-07
CPM10104	59.82	24.1	132.11	-12.46	0	35	2026-01-07
CPM10105	59.1	23.8	130.51	-12.31	0	35	2026-01-07
CPM10106	59.46	23.88	131.08	-12.17	0	35	2026-01-07
CPM10107	59.81	23.86	131.4	-11.78	0	35	2026-01-07
CPM10108	59.62	22.93	128.42	-9.18	0	35	2026-01-07
CPM10109	59.41	23.32	129.37	-10.56	0	35	2026-01-07
CPM10110	59.88	23.24	129.61	-9.85	0	35	2026-01-07
CPM10111	59.27	23.75	130.51	-11.97	0	35	2026-01-07
CPM10112	59.54	23.85	131.09	-12.02	0	35	2026-01-07
CPM10113	59.79	23.71	130.92	-11.34	0	35	2026-01-07
CPM10114	59.63	23.51	130.15	-10.89	0	35	2026-01-07
CPM10115	59.38	23.49	129.85	-11.09	0	35	2026-01-07
CPM10116	59.43	23.9	131.13	-12.26	0	35	2026-01-07
CPM10117	58.63	23.92	130.38	-13.11	0	35	2026-01-07
CPM10118	59.49	23.17	129.0	-10.02	0	35	2026-01-07
CPM10119	59.58	23.55	130.23	-11.06	0	35	2026-01-07
CRM10101	59.76	24.21	132.39	-12.87	0	35	2026-01-07
CRM10102	59.11	23.77	130.42	-12.2	0	35	2026-01-07
CRM10103	59.33	23.59	130.11	-11.44	0	35	2026-01-07
CRM10104	59.49	23.81	130.93	-11.94	0	35	2026-01-07
CRM10105	59.51	23.54	130.13	-11.1	0	35	2026-01-07
CRM10106	59.2	23.92	130.97	-12.57	0	35	2026-01-07
CRM10107	59.39	23.35	129.45	-10.68	0	35	2026-01-07
CRM10108	59.11	23.3	129.02	-10.8	0	35	2026-01-07
CRM10109	58.85	23.6	129.65	-11.95	0	35	2026-01-07
CRM10110	59.59	23.35	129.65	-10.48	0	35	2026-01-07
CRM10111	60.05	23.6	130.84	-10.74	0	35	2026-01-07
CRM10112	58.73	24.27	131.53	-14.06	0	35	2026-01-07
CRM10113	59.45	23.28	129.31	-10.4	0	35	2026-01-07
CRM10114	59.63	23.77	130.93	-11.67	0	35	2026-01-07
CRM10115	59.07	23.62	129.91	-11.78	0	35	2026-01-07
CRM10116	59.5	23.49	129.97	-10.96	0	35	2026-01-07
CRM10117	59.49	22.71	127.62	-8.64	0	35	2026-01-07
CRM10118	59.56	23.86	131.16	-12.03	0	35	2026-01-07
CRM10119	60.01	24.44	133.32	-13.3	0	35	2026-01-07
CSP10101	60.3	23.57	131.02	-10.41	0	35	2026-01-07
CSP10102	59.69	23.69	130.76	-11.37	0	35	2026-01-07
CSP10103	59.98	23.14	129.4	-9.44	0	35	2026-01-07

CSP10104	59.5	24.18	132.02	-13.03	0	35	2026-01-07
CSP10105	58.37	23.78	129.73	-12.98	0	35	2026-01-07
CSP10106	58.92	23.68	129.97	-12.13	0	35	2026-01-07
CSP10107	60.01	23.7	131.1	-11.08	0	35	2026-01-07
CSP10108	60.04	23.97	131.95	-11.88	0	35	2026-01-07
CSP10109	59.73	24.18	132.28	-12.82	0	35	2026-01-07
CSP10110	59.17	23.38	129.3	-10.95	0	35	2026-01-07
CSP10111	60.2	23.57	130.92	-10.52	0	35	2026-01-07
CSP10112	58.88	23.7	129.98	-12.22	0	35	2026-01-07
CSP10113	59.71	23.62	130.58	-11.16	0	35	2026-01-07
CSP10114	58.54	23.22	128.21	-11.12	0	35	2026-01-07
CSP10115	60.24	23.31	130.15	-9.68	0	35	2026-01-07
CSP10116	58.98	23.83	130.46	-12.5	0	35	2026-01-07
CSP10117	59.14	23.65	130.08	-11.81	0	35	2026-01-07
CSP10118	60.24	23.75	131.5	-11.02	0	35	2026-01-07
CSP10119	59.78	24.05	131.92	-12.37	0	35	2026-01-07
CSR10101	59.55	23.72	130.69	-11.6	0	35	2026-01-07
CSR10102	59.08	23.54	129.7	-11.53	0	35	2026-01-07
CSR10103	59.4	23.84	130.9	-12.11	0	35	2026-01-07
CSR10104	59.54	23.26	129.31	-10.22	0	35	2026-01-07
CSR10105	59.9	23.51	130.44	-10.64	0	35	2026-01-07
CSR10106	58.98	23.84	130.51	-12.54	0	35	2026-01-07
CSR10107	59.35	23.94	131.18	-12.47	0	35	2026-01-07
CSR10108	59.42	23.89	131.08	-12.24	0	35	2026-01-07
CSR10109	60.11	23.22	129.77	-9.55	0	35	2026-01-07
CSR10110	58.91	23.48	129.35	-11.52	0	35	2026-01-07
CSR10111	59.8	23.69	130.87	-11.27	0	35	2026-01-07
CSR10112	59.53	23.48	129.97	-10.91	0	35	2026-01-07
CSR10113	59.44	23.45	129.79	-10.9	0	35	2026-01-07
CSR10114	59.07	23.21	128.7	-10.57	0	35	2026-01-07
CSR10115	59.72	23.69	130.8	-11.35	0	35	2026-01-07
ITM10101	59.01	23.65	129.96	-11.94	0	35	2026-01-07
ITM10102	60.03	24.03	132.11	-12.05	0	35	2026-01-07
ITM10103	59.46	23.68	130.5	-11.58	0	35	2026-01-07
ITM10104	59.39	23.8	130.8	-12.02	0	35	2026-01-07
ITM10105	58.65	23.86	130.23	-12.94	0	35	2026-01-07
ITM10106	59.31	23.62	130.19	-11.56	0	35	2026-01-07
ITM10107	59.87	23.36	129.95	-10.22	0	35	2026-01-07
ITM10108	59.36	24.13	131.75	-13.02	0	35	2026-01-07
ITM10109	59.38	23.88	131.03	-12.26	0	35	2026-01-07
ITM10110	59.29	24.01	131.31	-12.73	0	35	2026-01-07
ITM10111	59.01	23.89	130.67	-12.64	0	35	2026-01-07
ITM10112	59.37	23.64	130.29	-11.56	0	35	2026-01-07
ITM10113	59.04	23.16	128.53	-10.45	0	35	2026-01-07
ITM10114	58.47	23.84	129.99	-13.05	0	35	2026-01-07
ITM10115	58.52	23.91	130.25	-13.21	0	35	2026-01-07
ITM10116	59.4	24.14	131.83	-13.03	0	35	2026-01-07
ITM10117	59.83	24.11	132.16	-12.5	0	35	2026-01-07
ITM10118	58.74	23.62	129.59	-12.11	0	35	2026-01-07
ITM10119	60.27	23.85	131.83	-11.3	0	35	2026-01-07
ORH10101	59.05	23.61	129.9	-11.79	0	35	2026-01-07
ORH10102	59.9	23.63	130.78	-10.99	0	35	2026-01-07
ORH10103	58.17	23.58	128.91	-12.57	0	35	2026-01-07
ORH10104	59.15	23.4	129.36	-11.07	0	35	2026-01-07
ORH10105	59.77	23.61	130.6	-11.07	0	35	2026-01-07
ORH10106	59.75	23.41	129.98	-10.47	0	35	2026-01-07
ORH10107	60.58	23.56	131.27	-10.11	0	35	2026-01-07
ORH10108	60.14	23.34	130.17	-9.88	0	35	2026-01-07
ORH10109	59.57	23.2	129.17	-10.03	0	35	2026-01-07
ORH10110	58.98	24.03	131.08	-13.12	0	35	2026-01-07
ORH10111	59.76	23.37	129.86	-10.35	0	35	2026-01-07
ORH10112	59.68	23.19	129.26	-9.9	0	35	2026-01-07
ORH10113	59.68	23.77	130.97	-11.62	0	35	2026-01-07
ORH10114	59.56	23.02	128.64	-9.51	0	35	2026-01-07
ORH10115	59.7	23.58	130.45	-11.05	0	35	2026-01-07
QSL10101	59.73	23.39	129.9	-10.44	0	35	2026-01-07
QSL10102	59.15	23.86	130.72	-12.42	0	35	2026-01-07
QSL10103	59.37	23.95	131.21	-12.47	0	35	2026-01-07
QSL10104	59.0	23.58	129.74	-11.74	0	35	2026-01-07
QSL10105	59.47	23.21	129.11	-10.16	0	35	2026-01-07
QSL10106	59.2	23.69	130.28	-11.88	0	35	2026-01-07
QSL10107	59.48	23.67	130.49	-11.52	0	35	2026-01-07
QSL10108	60.04	23.53	130.62	-10.54	0	35	2026-01-07
QSL10109	59.18	23.46	129.55	-11.2	0	35	2026-01-07
QSL10110	59.74	24.04	131.85	-12.37	0	35	2026-01-07

QSL10111	59.31	23.75	130.57	-11.96	0	35	2026-01-07
QSL10112	60.42	23.91	132.15	-11.32	0	35	2026-01-07
QSL10113	60.79	23.37	130.89	-9.31	0	35	2026-01-07
QSL10114	59.23	23.71	130.35	-11.88	0	35	2026-01-07
QSL10115	59.53	23.42	129.78	-10.72	0	35	2026-01-07
QSL10116	59.43	23.1	128.72	-9.86	0	35	2026-01-07
QSL10117	59.34	23.1	128.64	-9.95	0	35	2026-01-07
QSL10118	59.25	23.58	130.0	-11.5	0	35	2026-01-07
QSL10119	58.5	23.4	128.69	-11.69	0	35	2026-01-07
RTN10101	58.95	24.11	131.29	-13.38	0	35	2026-01-07
RTN10102	59.52	23.25	129.26	-10.22	0	35	2026-01-07
RTN10103	58.99	23.65	129.96	-11.97	0	35	2026-01-07
RTN10104	59.61	23.7	130.71	-11.49	0	35	2026-01-07
RTN10105	59.25	23.31	129.17	-10.67	0	35	2026-01-07
RTN10106	59.55	23.74	130.77	-11.67	0	35	2026-01-07
RTN10107	59.85	23.48	130.3	-10.59	0	35	2026-01-07
RTN10108	59.71	22.98	128.64	-9.23	0	35	2026-01-07
RTN10109	59.31	24.25	132.07	-13.44	0	35	2026-01-07
RTN10110	59.61	23.6	130.4	-11.18	0	35	2026-01-07
RTN10111	59.97	23.38	130.11	-10.17	0	35	2026-01-07
RTN10112	59.08	23.45	129.42	-11.26	0	35	2026-01-07
RTN10113	60.1	24.31	133.03	-12.83	0	35	2026-01-07
RTN10114	59.23	24.36	132.32	-13.87	0	35	2026-01-07
RTN10115	59.49	23.71	130.62	-11.64	0	35	2026-01-07
RTN10116	59.6	24.4	132.8	-13.59	0	35	2026-01-07
RTN10117	59.27	23.27	129.08	-10.55	0	35	2026-01-07
RTN10118	59.47	23.34	129.5	-10.55	0	35	2026-01-07
RTN10119	59.41	23.42	129.68	-10.86	0	35	2026-01-07

=== Customer_Module_model_summary.txt ===

NOTE ON THE SECTION BELOW: 'Overall Risk', the 'Risk Scenarios' ranking, and 'Follow-up Actions' are computed from randomly simulated stress-test scenarios (seeded for reproducibility), not from statistical evidence measured in this client's actual data. Treat them as illustrative stress-test projections, not as data-driven findings. (Model Evaluation Metrics further below ARE fit on the real data and are not affected by this note.)

Overall Risk: 0.3802845671230959

Risk Scenarios in Higher-First Priority:

Scenario_10: 0.9579666868632705

Scenario_9: 0.8037397818942802

Scenario_8: 0.638736195276424

Scenario_7: 0.48976899947419505

Scenario_6: 0.3611336711829887

Scenario_5: 0.25170614480786363

Scenario_4: 0.1591669199583243

Scenario_3: 0.09014483389627911

Scenario_2: 0.03943448241278718

Scenario_1: 0.011047955464545458

Average MSE across models is 9.79311356752109e-06, indicating strong confidence in the risk assessment.

Follow-up Actions:

- High risk in Scenario_10 (Avg Risk=0.958) driven by metrics: CHG10107, CRM10109, BIH10107. Suggested mitigation: prioritize monitoring and corrective actions for these metrics.
- High risk in Scenario_9 (Avg Risk=0.804) driven by metrics: CFS10101, CBC10110, CSP10101. Suggested mitigation: prioritize monitoring and corrective actions for these metrics.

- High risk in Scenario_8 (Avg Risk=0.639) driven by metrics: CFS10102, CFS10108, CIM10109. Suggested mitigation: prioritize monitoring and corrective actions for these metrics.

Metrics involved in High-Risk Scenarios:

Scenario_10 (Avg Risk=0.958):

- CHG10107: Charge Reversal Rate
- CRM10109: Customer Engagement Rate
- BIH10107: Inquiry Accuracy Rate

Scenario_9 (Avg Risk=0.804):

- CFS10101: Response Rate
- CBC10110: Customer Satisfaction (CSAT) with Balance Management
- CSP10101: Segmentation Accuracy

Scenario_8 (Avg Risk=0.639):

- CFS10102: Customer Satisfaction (CSAT) Score
- CFS10108: Sentiment Analysis
- CIM10109: Data Usage Rate

Baseline Predictor (Mean of CUST_001):

MSE: 1.5311976057378582

R-squared: 0.0

RMSE: 1.2374156964164704

MAPE: 1.663849335374603%

PyTorch Model Evaluation:

MSE: 0.0009623768553458673

R-squared: 0.9993714874868276

RMSE: 0.031022199395688684

MAPE: 0.0403930972835058%

Model Evaluation Metrics:

MSE: 0.0009623768553458673

RMSE: 0.031022199395688684

R²: 0.9993714874868276

MAPE: 0.0403930972835058%

Risk Confidence Score: High

Recommendation - Continuously monitor performance metrics and adjust strategies as needed.

Recommendation - Reassess risk management plans periodically to ensure effectiveness.

-----End of Customer Module Risk Analysis Summary-----